

Monitoring Corruption and Overcoming the Collective Action Problem*

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Abstract

Recent corruption scholarship emphasizes that monitoring consistent with the principal-agent model mostly fails in the presence of systemic corruption. Yet incremental monitoring sometimes works. We explain why using a simple collective-action model with multiple equilibria and an assurance-game structure. It captures the limits of top-down supervision under systemic corruption while specifying the citizen-level payoffs, beliefs, and strategic reasoning that can sustain or shift equilibria. Instead of treating monitoring payoffs as monolithic, we separate them into process-level collective benefits and actor-level private benefits. Both can lower the threshold for taking costly action against corruption, but private benefits are more effective per unit because collective benefits also increase the payoff to free-riding. Overall, our model enables collective-action accounts of corruption to better explain incremental change and heterogeneous outcomes, and our monitoring benefits distinction provides a policy design heuristic.

Keywords: corruption, monitoring, collective action, principal-agent theory

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The abuse of public office or entrusted power for private gain has a common name: corruption.¹ Despite considerable efforts to monitor and sanction corrupt officials in line with the principal-agent model, corruption persistently afflicts both rich and poor countries (Findley, Nielson and Sharman, 2014; Fukuyama and Recanatini, 2021). A key reason is that especially when corruption is systemic, there is often a shortage of credible monitors or “principled principals” who are willing to depart from social norms and address corruption (e.g., Peiffer and Alvarez, 2016).² Yet monitoring sometimes works. Why?

In this paper, we explain how monitoring and other more incremental interventions can be effective at reducing corruption. To do so, we build on Persson, Rothstein and Teorell (2013, 457) and Fisman and Golden (2017, 6) and advance a multiple-equilibria model of corruption. By embedding monitoring into a collective action model, we capture the risk of “unprincipled principals”, citizens’ incentives, and some functional benefits corruption may provide in some settings (see Marquette and Peiffer, 2018). Given the model’s assurance-game structure, multiple outcomes are possible, and the game does not devolve into a pre-determined, single-equilibrium prisoner’s dilemma or game of harmony.³

Existing accounts of corruption as a collective action problem excel at identifying why corruption persists, but our model goes further by also explaining shifts from high- to low-corruption equilibria. To that end, we incorporate Schelling’s (1978) fundamental insight from focal points and tipping games: collective action depends on contingent behavior (Fisman and Golden, 2017, 5). We operationalize contingent behavior, and how it varies across individuals, through beliefs about strategic reasoning and others’ willingness to take costly action against corruption. Then, we apply Medina’s (2007) variant of Harsanyi and Selten’s (1988) tracing procedure for equilibrium selection. It enables the model to show the conditions under which monitoring can shift the threshold separating high- and low-corruption

¹See Rose-Ackerman and Palifka (2016, Chapter 1) for more on the various definitions of corruption.

²For related earlier work, see Andvig and Moene (1990), Aidt (2003), and Kingston (2008). For related empirical work documenting how rule-breaking is contagious and generally higher in corrupt societies, see Gächter and Schulz (2016) and Shalvi (2016).

³See Appendix D for an overview of basic collective action games.

equilibria. Like [Stephenson \(2020\)](#), our approach does not require a coordinated, big-bang reform (see [Rothstein, 2011a](#)). Unlike [Stephenson \(2020\)](#), however, our model retains multiple equilibria emphasized by collective-action accounts.

Substantively, a key contribution of our model entails distinguishing between collective and private citizen-level monitoring benefits. Collective benefits are non-excludable: once at least one citizen takes costly action against corruption, the deterrence, detection, or sanctioning benefits from monitoring can accrue to cooperators and free-riders alike. That leakage or spillover property characterizes audits, e-procurement systems, and other process-centered mechanisms generating collective benefits. By contrast, private benefits are excludable: whistleblower protections, ombuds remedies, and other forms of actor-specific support only accrue to citizens who take action against corruption. To be clear, institutions facilitating either collective or private benefits may be subject to “unprincipled principals” and credibility problems ([Ostrom, 1990, 17](#); [Rothstein, 2021, Ch. 7](#)). Nevertheless, they need not share the same norms, incentives, or constraints as the actors embedded in the setting where corruption occurs (e.g., [Bersch, Praça and Taylor, 2017](#)). In this light, level-of-analysis distinctions and recognition of within-society heterogeneity can guide reform efforts even in the presence of highly systemic corruption.

Understanding the incidence of monitoring benefits also reveals an asymmetry that matters for policy design. Given that collective benefits leak to free-riders but private benefits do not, our model shows that equally-sized private benefits are more effective per unit at lowering the threshold for taking costly action against corruption. That does not imply private benefits are universally more effective. Private benefits resemble Olson-style selective incentives, but with a crucial difference: unlike [Olson’s \(1965\)](#) prototypical example of paying labor union dues, citizens receiving private benefits must take costly action themselves. Another consideration is that private, actor-centric benefits, such as legal support from NGOs, are not appropriate for every situation. Sometimes, collective, process-oriented benefits from transparency initiatives or citizen report cards are larger, cheaper, more scalable, or more

credible. Regardless of the particularities of the context, the collective and private benefits distinction allows our collective action model to better capture anti-corruption tools and oversight measures commonly associated with principal-agent logic (e.g., [Olken, 2007](#); [Ferraz and Finan, 2008](#); [Marquette and Peiffer, 2018](#); [Bersch, 2019](#); [Lagunes, 2021](#)).

1. Corruption as a Principal-Agent Problem

The most common definition of corruption is the “misuse of public office [or entrusted power] for private gain” (e.g. [Treisman, 2000](#); [Transparency International, 2009](#)).⁴ The principal-agent model, along with most monitoring efforts, follows directly from that definition ([Lambsdorff, 2002](#), 97).

There are two basic manifestations of the principal-agent model. Under the first, the public (the principal) delegates power to a politician (the agent) by voting them into office. Under the second manifestation, a politician or highly-ranked bureaucrat (the principal) delegates power to lower-ranking bureaucrats (the agents). Under both manifestations, there are potential efficiency gains from delegation, assuming that agents do not misuse their power for private gain. To prevent it from taking place, principals must monitor and sanction corrupt agents ([Klitgaard, 1988](#); [Vaubel, 2006](#)).

Incentives help underscores why principal-agent approaches to corruption often fail. On the principal side, “principled principals” are in short supply when corruption is systemic ([Aidt, 2003](#); [Persson, Rothstein and Teorell, 2013](#); [Peiffer and Alvarez, 2016](#)). On the agent side, it is frequently not within agents’ self-interests to take action against corruption ([Bicchieri and Ganegoda, 2017](#), 180). Underpinning both principals’ and agents’ behaviors are shared norms, expectations, beliefs, and other informal institutions. They tend to be stronger

⁴As [Dixit \(2016\)](#) explains, this definition of corruption maps well onto bribery, which entails a supply side (i.e., those providing the bribes—the private sector) and a demand side (i.e., those accepting/requesting the bribes—the public sector). In reality, though, corruption entails much more than bribery. For example, corruption entails kickbacks, coercion/extortion, nepotism, cronyism, financial fraud, electoral fraud, collusion, obstruction, and patronage ([Søreide, 2014](#)). For more on the definition of corruption, refer to [Rose-Ackerman and Palifka \(2016\)](#) and [Fisman and Golden \(2017\)](#).

than any formal institutions designed to control corruption. That is why monitoring-based anti-corruption reforms following the principal-agent model often fail when corruption is systemic (Collier, 2000; Persson, Rothstein and Teorell, 2013; Fisman and Golden, 2017).

2. Corruption as a Collective Action Problem

A collective action problem “arise[s] when the individual pursuit of self-interest generates socially undesirable outcomes” (Ferguson, 2013, 4). Corruption is a collective action problem because most people in corrupt societies would benefit from having less corruption. By the same token, resisting corruption—and contributing to the public good—is not in most people’s individual self-interest unless citizens expect others to do so as well. That is not just the case for the select people who financially benefit from corruption but also for its victims. Taking any type of action against corruption, such as refusing to pay a bribe, can carry potential costs such as intimidation, violence, inability to obtain government services, and being put on a blacklist (e.g., Kingston, 2008; Wrong, 2009).

Among the standard collective action games (see Appendix D), the prisoner’s dilemma and assurance game provide the most compelling macrostructural manifestations of corruption as a collective action problem (Yap, 2013; Dixit, 2018).⁵ In their most basic forms, both the prisoner’s dilemma and the assurance game involve two players. Each player’s essential task is to (simultaneously) make a decision about how to behave without being able to coordinate with the other side. In the case of corruption, the preferred collective decision is to reject it, but it becomes privately optimal to accede and tolerate corruption when the other side has strong incentives to do so as well.

Although the prisoner’s dilemma elucidates how citizens may want to free-ride on others’

⁵Ostensibly, the game of harmony does not describe corruption well in corrupt countries, because incentives to defect and thus free-ride on the contributions of others are still present; otherwise, corruption would not be so prevalent across the world. Deadlock is also inappropriate, since the payoffs of defecting are not that high for all citizens; if so, there would be no reason to mitigate corruption. The game of chicken is similarly unconvincing for describing corruption: taking action is the challenge, and chicken assumes that someone always takes action. Notably, there are two pure strategy Nash equilibria in Chicken. See Appendix D.

actions against corruption, the model has a significant flaw that makes it less useful for describing corruption than an assurance game. Because the prisoner’s dilemma necessarily entails a dominant strategy and unique equilibrium at defection (see Appendix D), it cannot explain variation in the outcome of whether citizens take action against corruption (Persson, Rothstein and Teorell, 2013). By contrast, the assurance game can explain such variation because of its multiple equilibria: one pure strategy Nash equilibrium at both parties taking action ($PSNE_{Cooperate}$), another at doing nothing ($PSNE_{Defect}$), and a mixed-strategy Nash equilibrium (MSNE).⁶

The stock assurance game, however, is insufficient to explain how citizens’ calculations about whether to take action against corruption can change. Fisman and Golden (2017, 5) correctly suggest that it relates to Schelling’s (1978) fundamental insight from focal points, tipping games, and threshold models on contingent behavior: that is, a person’s decision to take action against corruption depends on her belief about whether others will reciprocate (Dong, Dulleck and Torgler, 2012; Lee and Guven, 2013; Gneezy, Saccardo and van Veldhuizen, 2019; Sundström, 2019).⁷

A main drawback of focal points and tipping games is that expectations and other determinants of contingent behavior vary with events in the external environment (Medina, 2007, 59-61). For example, even social trust, which numerous authors argue is the most important determinant of corruption (e.g., Rothstein, 2021), changes according to a new leader taking power, repression, scandals, etc. By construction, therefore, prediction is difficult when focal points and tipping games have multiple equilibria (Medina, 2007, 60; Medina, 2018, 47).

To address these challenges, we shift from identifying possible equilibria to analyzing equilibrium selection and stability. Our approach uses a variant of Harsanyi and Selten’s (1988) tracing procedure to vary the weight citizens place on strategic reasoning versus initial beliefs about others’ likely behavior. By doing so, we can show when monitoring-generated

⁶See Dixit, Skeath and Reiley (2014) and Humphreys (2017).

⁷For a threshold model of revolution, see Kuran (1989).

benefits make costly action against corruption more likely without turning the assurance game into one of harmony or eliminating the high-corruption equilibrium altogether. In this way, we follow [Medina's \(2007, 7\)](#) layperson's theory of collective action:

“When individuals can achieve some beneficial result by coordinating in a group, they are likely to coordinate. As the potential benefits of coordination increase (or the costs decrease), these individuals are more likely to coordinate and, conversely, as the potential benefits decrease (or the costs increase), they are less likely to do so.”

3. A Model of Corruption as a Collective Action Problem

We model collective action against corruption as a one-shot assurance game between two strategic and representative citizens, i and j , as well as an unstrategic bureaucrat, who always solicits a bribe. The bureaucrat's incentives still enter the model via citizen decisions about retaliation costs, and by having the bureaucrat always request a bribe, we isolate the citizen-level collective action problem under fully systemic corruption. That makes our game illustrative of a hard test for a hypothesis that monitoring is not futile in such settings.

Let Γ^A denote the baseline game that we use as a point of departure for subsequent games Γ^B , Γ^C , and Γ^D . All of these games follow [Persson, Rothstein and Teorell's \(2013\)](#) account of systemic corruption. Citizens prefer a low-corruption outcome if others also cooperate. However, prevailing norms make corruption the default and unilateral cooperation costly. By contrast, defection has none of these costs but contributes nothing to the public good. For the sake of simplicity, we represent the costly actions as refusing to pay a bribe or reporting a corrupt bureaucrat, but our model can apply to other anti-corruption efforts as well. Because we aim to isolate how monitoring changes payoffs when citizens act simultaneously and cannot observe or coordinate with one another, our model does not incorporate

group-size thresholds, iteration, or multiplayer structures.⁸ Our model thus does not constitute a hidden-action/information principal-agent game or feature signaling or updating of beliefs.⁹

3.1. The Basic Setup: Collective Action without Monitoring

Keeping with Medina’s (2007) notation for collective action games, we denote payoffs for player i as W_1 when both sides cooperate; W_2 where player i defects but player j cooperates; W_3 where player i cooperates but player j defects; and, finally, W_4 where both players defect. Consistent with traditional assurance game payoffs,¹⁰ and keeping all component parameters non-negative in line with Equation (B.2), we define W_1 through W_4 such that:

- W_1 : If and only if citizen i refuses to pay the bribe (i.e., she “cooperates”), and another citizen j joins her in doing so. By refusing to pay the bribe, she (most likely) forgoes the opportunity to obtain the demanded service, but provides a public good. As citizen j matches her behavior, the benefits from the public good become more pronounced ($u_{\text{public good}} = t$). She does, however, face the risk of retaliation from the bureaucrat, for which she pays a cost, r . Mathematically, her payoff is: $W_1 \equiv u_i(C_i|C_j) = t - r > 0$.
- W_2 : If and only if citizen i pays the bribe (i.e., she “defects”), but citizen j opts to cooperate, thus providing for a corruption-fighting public good that citizen i can free-ride on. By citizen i not supplying the public good herself, its benefit is reduced ($u_{\text{public good}} = s, t > s > 0$). By paying the bribe, however, citizen i virtually guarantees receipt of the service, which she values at g . The (immediate financial) cost of her bribe is denoted by b . In line with functionalist scholarship on corruption (Marquette and Peiffer, 2018), she also avoids the problem of retaliation from the bureaucrat or anyone

⁸Substantively, corruption is a clandestine behavior that provides a very low signaling environment to facilitate citizens’ short-term preference falsification (see Kuran, 1991). Additionally, group size thresholds for overcoming corruption are context-dependent, and relevant collective action scholarship offers no firm guidance as well (Sandler, 2015; Weimann et al., 2019).

⁹For models along these lines, see Tirole (1986), Baliga (1999), Laffont and Martimort (1999), Besley (2006), Ryvkin and Serra (2012), and Brandt and Svendsen (2013).

¹⁰Readers unfamiliar with payoff structures of basic collective action games may refer to Appendix D.

else by paying the bribe. Hence, citizen i 's payoff is: $W_2 \equiv u_i(D_i|C_j) = s + g - b > 0$.

- W_3 : If and only if citizen i cooperates by refusing to pay the bribe, thus investing into the public good, but her efforts are not matched by citizen j . As previously noted, refusal to pay results in foregone reception of services and possible retaliation from the bureaucrat requesting the bribe. By singularly investing in the public good, her benefits from public good are small (s). In total, the action yields: $W_3 \equiv u_i(C_i|D_j) = s - r$.
- W_4 : If and only if neither citizen i nor citizen j cooperate, no investment in the public good occurs. More concretely, both citizens receive their services in return for paying the bribe. To normalize the game at zero, we assume a functionalist status-quo outcome that produces no individual net gain or loss for either citizen i or others represented by j , where the bureaucrat can extract the exact value of the good: $W_4 \equiv u_i(D_i|D_j) = g - b = 0$.¹¹

For version Γ^A to be an assurance game in line with Persson, Rothstein and Teorell (2013), first, the potential costs of retaliation must exceed those of singular cooperation (i.e., $r > s$). Second, despite the threat of retaliation, the societal benefits of successful collective action must eclipse the payoff from singular defection (i.e., $W_1 > W_2$, or $t - r > s + g - b$).¹² Taken together, any of the games covered here are assurance-type games if and only if $W_1 > W_2 > W_4 > W_3$. As a result, Γ^A through Γ^D must produce three equilibria: Two pure strategy Nash equilibria ($PSNE_{Cooperate}$ and $PSNE_{Defect}$) and one mixed-strategy Nash equilibrium (MSNE, denoted by α^*). As we show in Appendix B.1.1, the existence of the PSNEs is contingent on these aforementioned assumptions. For version Γ^A , as shown in Appendix B.1.2, the MSNE identifies the cooperation threshold at which citizens are

¹¹Any potential bargain over the size of the bribe amount given the value of the provided good depicts a hierarchical divide-the-dollar type of interaction between *strategic* bureaucrats and citizens. As stated earlier, the bureaucrats in this game are nonstrategic. In our model, centering $g - b = 0$ does not imply that bribery is harmless but normalizes the status quo payoff in the face of systemic corruption.

¹²For some, this assumption might be considered a strong one. Indeed, forgone services might be at the forefront of many citizens' thoughts as they contemplate the decision to refuse a bribe, despite the undeniable citizen benefits of systemically fighting corruption. However, without this assumption, the game defaults to a prisoner's dilemma. As we emphasize earlier, the prisoner's dilemma does not adequately describe the problem of corruption.

indifferent between taking costly action and doing nothing, denoted by α_i (or α_j), where:

$$\alpha_i^{*,A} = \frac{g - b + r - s}{t - 2s} \quad (1)$$

More simply, given that we normalize $g - b = 0$ to be the status-quo payoff (see above):

$$\alpha_i^{*,A} = \frac{r - s}{t - 2s} \quad (2)$$

As [Medina \(2007, 147\)](#) shows, we can represent the MSNE more generally as:

$$\alpha_i^* = \frac{W_3 - W_4}{W_2 - W_4 + W_3 - W_1} \quad (3)$$

3.2. The Augmented Setup: Collective Action with Monitoring

Building on the basic setup above, we introduce two functionally distinct sets of citizen-level monitoring benefits: collective and private. Both can increase the likelihood of costly anti-corruption action, but they do so through different payoff channels. The collective versus private distinction does not hinge on whether monitoring originates inside or outside the organization where corruption occurs. Instead, the distinction hinges on the incidence of the resulting benefits: that is, whether they spill over to citizens who do not act or accrue only to those who do. We explain each channel in turn and then examine the case in which both collective and private monitoring benefits are present.

3.2.1. Collective Benefits of Monitoring

The collective benefits of monitoring (c) are non-excludable: once at least one citizen takes costly action against corruption, the deterrence, detection, or sanctioning benefits from monitoring can accrue to cooperators and free-riders alike. The non-excludable property of c stems from the fact that it works through the process in which corruption occurs rather than through the individual actor alone. When monitoring makes one citizen's refusal, complaint,

or report consequential, it can reduce corruption in the broader process, generating a positive process externality for others transacting within the same organization or setting.

Many types of monitoring instruments can generate c . Examples include supervisory oversight of bureaucrats; internal audits; routine external audits by a supreme audit institution or private firms; e-procurement and e-financial management systems; citizen report cards; and social audits. In line with [Scott \(1998\)](#), these monitoring instruments make corruption more legible by generating evidentiary records of an otherwise clandestine phenomenon.

Legibility, however, does not guarantee accountability. Monitoring instruments generate c only insofar as they make a citizen's refusal, complaint, or report more likely to trigger detection, reversal, sanctioning, or broader deterrence. Citizens may therefore perceive collective benefits as small when monitoring institutions remain vulnerable to "unprincipled principals" or weak bureaucratic professionalism. In this sense, our operationalization of c captures both the monitoring capacity in a given setting and the way the larger social contract shapes citizens' expectations about whether that capacity matters ([Rothstein, 2021](#)).

Formally, let Γ^B denote the game introducing c (see [Appendix B.2](#) for the normal form game). In Γ^B , c increases W_1 , W_2 , and W_3 , whereas W_4 remains constant due to lack of citizen cooperation. The increase in W_2 reflects the non-excludable nature of collective benefits: a citizen who defects can still benefit when another citizen's costly action activates c . Maintaining the assurance-game structure requires $s - r + c < 0$,¹³ and keeping with the normalization of $g - b = 0$, the MSNE becomes (see [Appendices B.2.1](#) and [B.2.2](#) for proofs):

$$\alpha_i^{*,B} = \frac{r - s - c}{t - 2s - c}, \quad \text{where } c > 0, \alpha_i^{*,B} < \alpha_i^{*,A}. \quad (4)$$

¹³Note that the assumption of $s - r + c < 0$ is more stringent than $s - r < 0$, because $c > 0$.

3.2.2. Private Benefits of Monitoring

In contrast to collective benefits, the private benefits of monitoring (p) accrue only to citizens who take costly action against corruption. These benefits are private because they are excludable, not necessarily because they derive from outside the organization where corruption occurs. It is also useful to clarify that p is not a typical [Olson \(1965\)](#) selective incentive: p is action-contingent, so it is more costly than, for example, paying membership dues and letting others take action.

Mechanisms generating p include whistleblower protections; insulated reporting or resolution channels (e.g., ombudsmen); and legal and other support from NGOs, international organizations, inspectors general, and other accountability actors. Of course, the actors facilitating p may face their own “unprincipled principals” and credibility problems ([Ostrom, 1990, 17](#); [Rothstein, 2021, Ch. 7](#)). Even so, their norms, incentives, and constraints often differ from those prevailing where the corrupt transaction occurs. That is a key reason why unbundling monitoring benefits by incidence is useful. Otherwise, there is a risk of obscuring underlying level-of-analysis heterogeneity (see [Gingerich, 2013b](#)).

Formally, let Γ^C represent the game introducing p , which increases W_1 and W_3 but does not affect W_2 and W_4 (see Appendix [B.3](#) for the normal form game). The payoff structure reflects the excludable nature of private benefits, as only cooperators receive the benefits. As with collective benefits, we do not assume that private protection fully solves corruption or eliminates the high-corruption equilibrium. Maintaining the structure of the assurance game, and centering around $W_4 = 0$, $s - r + p < 0$, the MSNE is (see Appendices [B.3.1](#) and [B.3.2](#) for proofs):

$$\alpha_i^{*,C} = \frac{r - s - p}{t - 2s}, \quad \text{where } p > 0, \alpha_i^{*,C} < \alpha_i^{*,A} \quad (5)$$

3.2.3. Collective and Private Benefits of Monitoring

Finally, consider the scenario in which there are both collective and private monitoring benefits. In most real-world settings, both forms of monitoring benefits exist. Bureaucracies usually contain supervisory hierarchies, audit systems, or procurement rules that can generate c . Many states and societies also contain horizontal, diagonal, or firewalled accountability institutions that can generate p . The key question is not whether these mechanisms exist on paper, but whether citizens perceive them as credible enough to alter their expected payoffs.

There are also instances in which the same organization or transaction can generate both collective and private benefits. Take, for example, a citizen reporting to an integrity office about a rigged procurement process. If the integrity office investigates, creates an evidentiary record, reverses the decision, sanctions officials, or deters future manipulation, it generates c because other actors can benefit even if they did not complain. If the integrity office also protects the complainant's identity, shields her from retaliation, offers legal support, or provides an individualized remedy, it generates p because the benefits accrue specifically to the actor taking the costly action.

Formally, let Γ^D denote the game with both collective and private monitoring benefits (see Appendices B.4-B.4.1). Under Γ^D , W_1 and W_3 increase by $c + p$, whereas W_2 increases by c , and W_4 remains unchanged when compared to Γ^A . To maintain the assurance-game structure, we assume $r - s > c + p$. Solving for the MSNE and normalizing around $W_4 = g - b = 0$, we obtain (see Appendix B.4.2):

$$\alpha_i^{*,D} = \frac{r - s - p - c}{t - 2s - c}, \quad \text{where } \forall p, c > 0: \alpha_i^{*,D} < \alpha_i^{*,B} < \alpha_i^{*,A} \text{ and } \alpha_i^{*,D} < \alpha_i^{*,C} < \alpha_i^{*,A} \quad (6)$$

Equation (6) shows that collective and private benefits are additive: combining them lowers the cooperation threshold more than either one alone. Nevertheless, comparing Γ^B and Γ^C also reveals an asymmetry of incidence. Given that c leaks to W_2 but p does not, Appendix B.6 shows that, under the assumptions necessary to maintain the assurance game,

p reduces the cooperation threshold more than c whenever:

$$p > \frac{c(t - s - r)}{t - 2s - c} \quad (7)$$

The larger takeaway is that private benefits need not exceed collective benefits in absolute size to have a larger effect on the cooperation threshold. When citizens value collective and private benefits equally, private benefits lower the threshold more because they attach only to cooperators rather than also increasing the payoff to free-riding.

4. Comparative Statics and Best Responses

The preceding subsections derive each game's threshold MSNE, α_i^* , which in this section we use to compute comparative statics for the monitoring parameters, c and p . Appendix C provides the proofs for the fully-augmented game with both types of monitoring, Γ^D . Given that Γ^A , Γ^B , and Γ^C reflect special cases of Γ^D , in which c and/or p are zero, we do not repeat the relevant proofs for the other games.

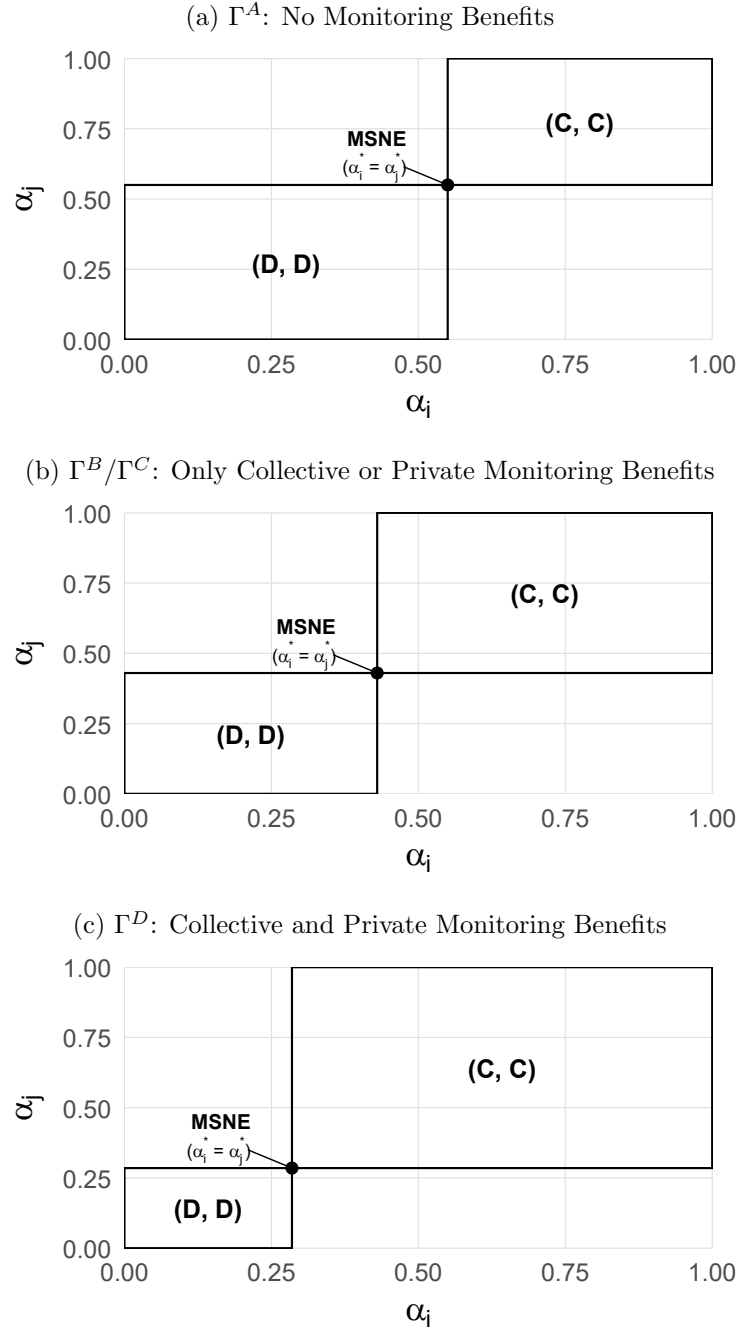
Deriving α_i^* with respect to the collective monitoring benefits, c , we obtain:

$$\frac{\partial \alpha_i^*}{\partial c} = \frac{r + s - p - t}{(t - 2s - c)^2} \quad (8)$$

Because the assurance-game assumptions imply $t > r + s$, and given that p and c are positive, the partial derivative of α_i^* must be negative. Hence, an increase in collective benefits emanating from monitoring coincides with a shrinking MSNE. Substantively, c does not create “principled principals” but changes the expected return to costly anti-corruption action.

The comparative statics for private monitoring benefits follow a similar logic. Recall that $t - 2s - c > 0$ is necessary to achieve a valid mixing strategy and, in turn, maintenance of all three assurance game equilibria for games Γ^B and Γ^C (see Appendices B.2.2 and B.3.2).

Figure 1: Simulated Best-Response Plots



Note: Each panel plots the best-response regions for citizens i and j . The lower-left region corresponds to mutual defection (D, D) , while the upper-right region corresponds to mutual cooperation (C, C) . The amounts of mutual cooperation and mutual defection in Γ^A are not the same in every context, so Figure 1a merely represents a simulated baseline with which to compare the other games. Figure 1b represents the general threshold-lowering effect of introducing either collective or private monitoring benefits. Figure 1b does not imply that Γ^B and Γ^C have identical thresholds, as Section 3.2 shows that equal-sized private benefits lower the threshold more because p does not increase the payoff to free-riding. Conditional on maintaining the assurance-game assumptions, Figure 1c shows that adding both c and p lowers the threshold more than adding either one alone.

Consequently, we can derive α_i^* with respect to p , obtaining:

$$\frac{\partial \alpha_i^*}{\partial p} = -\frac{1}{t - 2s - c} \quad (9)$$

Again, private benefits emanating from monitoring coincide with a shrinking threshold under the MSNE. While Appendix B.6's result holds, both collective and private monitoring benefits lower the threshold that separates cooperative and defecting basins of attraction. In other words, the result is not that monitoring eliminates the high-corruption equilibrium. It is that credible monitoring benefits reduce how optimistic citizens must be about others' cooperation before taking action themselves.

The comparative statics also yield a simple best-response interpretation (see Figure 1). For each citizen, the MSNE marks the threshold at which costly action against corruption becomes optimal. Below this threshold, defection is the best response; above it, cooperation is the best response. In the joint strategy space, the two citizens' cutoffs divide the unit square into regions of mutual defection, mutual cooperation, and asymmetric best responses. The lower-left and upper-right regions foreshadow the high- and low-corruption basins of attraction that the next section formalizes with stability sets. The unlabeled off-diagonal regions do not define basins on their own. As monitoring benefits lower the MSNE in Figures 1b and 1c, they shrink the mutually-reinforcing defection region and expand the mutually-reinforcing cooperation region.

5. Equilibrium Selection and Basins of Attraction

While the preceding section shows that monitoring can shift the threshold between basins of attraction without transforming the assurance game into one of harmony, this section shows why these incremental shifts matter for equilibrium selection. Our approach uses stability sets (Medina, 2007). Like global games (Carlsson and van Damme, 1993), stability sets evaluate equilibria by examining their robustness to alternative starting points

for how players reason. Global games rely on noisy signals and study how equilibria behave as signal noise vanishes. Stability sets instead begin with players' initial beliefs about others' behavior and then vary the weight players place on strategic reasoning. To map initial beliefs and strategic reasoning weights into basins of attraction consistent with a game's payoffs and equilibria, stability sets use a variant of [Harsanyi and Selten's \(1988\)](#) tracing procedure. Such an approach is useful for studying anti-corruption reforms given that they usually do not immediately change preferences or eliminate formal equilibria. Instead, reforms can change which equilibrium citizens expect others to help sustain.

As a baseline to understand the stability set, consider the standard utility function for any game Γ^0 with multiple equilibria under full strategic rationality or, more precisely, common-knowledge rationality ($\lambda = 1$):¹⁴

$$u_i^{0,\lambda=1}(\alpha_i, \alpha_j) = \alpha_i(\alpha_j W_1 + (1 - \alpha_j) W_3) + (1 - \alpha_i)(\alpha_j W_2 + (1 - \alpha_j) W_4) \quad (10)$$

Under Equation (10), citizen i 's expected utility depends on citizen j 's actual mixed strategy, α_j . However, that assumption is often too strong for understanding the collective action problem in the context of corruption. Citizens confronting corruption usually do not observe others' decisions, cannot coordinate with them, and cannot know with certainty whether others will resist or accommodate corruption.

To address these real-world constraints, we introduce beliefs about others' behavior. Formally, let β_j denote citizen i 's initial belief that citizen j will cooperate. In sociological terms, β_j represents a descriptive norm or local empirical expectation rather than an injunctive norm ([Bicchieri, 2017](#); [Kubbe, Baez-Camargo and Scharbatke-Church, 2024](#); [Simpser, 2025](#), 1004). In practice, the same institutional environment that affects the credibility of monitoring may also shape citizens' beliefs about others' behavior. Our model brackets this upstream process. By treating β_j as an initial condition, not as an outcome of monitoring,

¹⁴In game theory, common-knowledge rationality means that citizen i believes citizen j is rational; citizen j believes that citizen i is rational; citizen i believes that citizen j believes that citizen i is rational, etc.

the model allows us to isolate the strategic channel through which monitoring changes citizens' payoffs. This modeling choice does not require empirical independence between beliefs and monitoring, as it only holds initial beliefs fixed while examining how monitoring alters incentives within the game. Because the game remains simultaneous and citizens do not observe one another's choices, β_j does not represent a Bayesian posterior generated within the game. Instead, β_j follows a continuous distribution on $(0, 1)$ with probability density function $f(\beta_j)$ and cumulative distribution function F_{β_j} .

Following Medina's (2007) adaptation of Harsanyi and Selten's (1988) tracing procedure, we can now define the stability set of a strategy (σ) as the region of initial beliefs for which that strategy is a best response (BR). The stability set of cooperation is:

$$\sigma_i(\text{Cooperate}_i) = \{\beta_j \in [0, 1] : \text{Cooperate}_i \in BR_i(\beta_j)\} \quad (11)$$

Similarly, the stability set of defection is:

$$\sigma_i(\text{Defect}_i) = \{\beta_j \in [0, 1] : \text{Defect}_i \in BR_i(\beta_j)\} \quad (12)$$

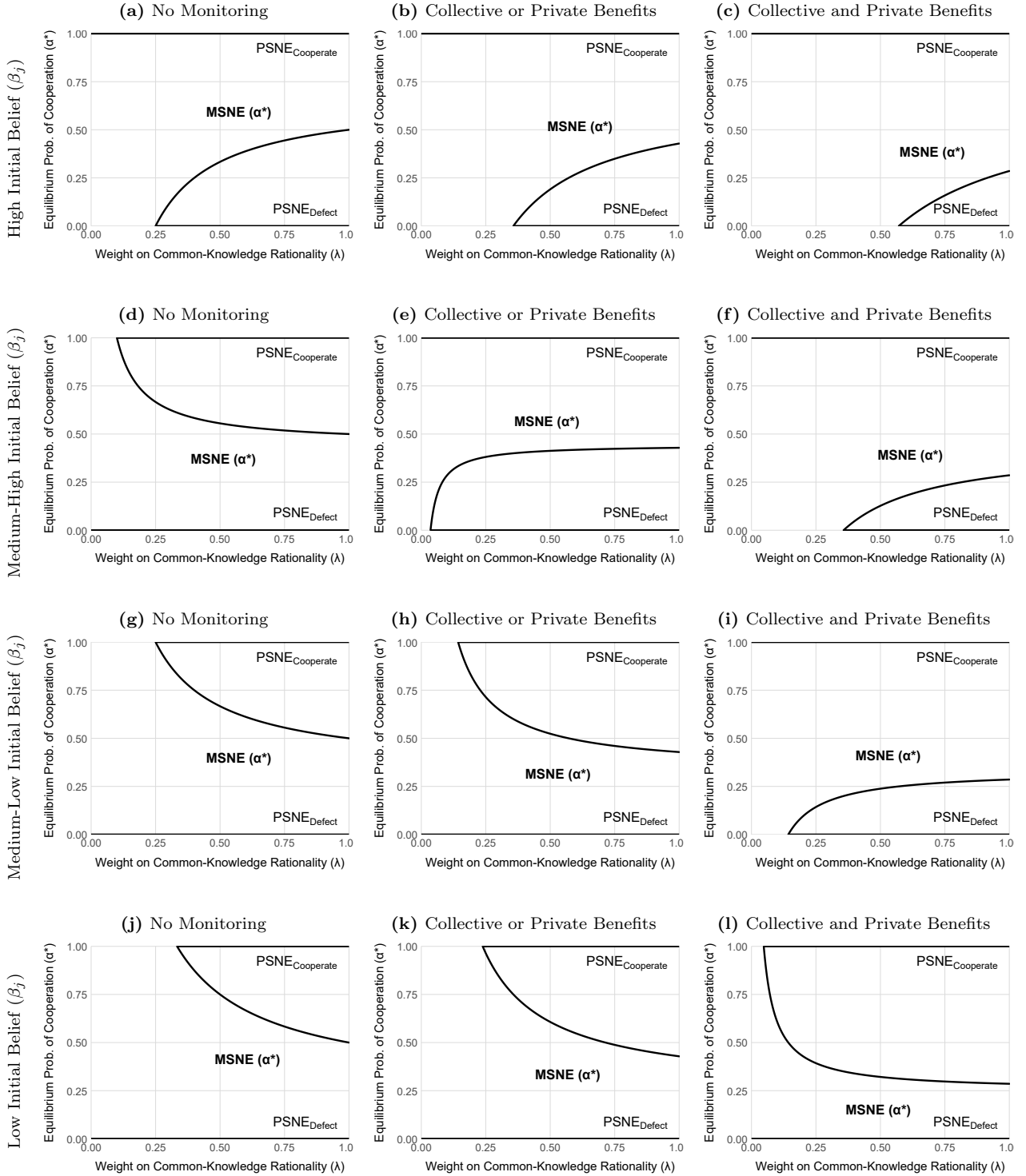
To capture how initial beliefs shape equilibrium stability, now consider citizen i 's weighted expected-utility function (see Appendix B.5):

$$\begin{aligned} u_i^{0,\lambda}(\alpha_i, \alpha_j, \beta_j) = & \lambda [\alpha_i(\alpha_j W_1 + (1 - \alpha_j)W_3) + (1 - \alpha_i)(\alpha_j W_2 + (1 - \alpha_j)W_4)] \\ & + (1 - \lambda) [\alpha_i(\beta_j W_1 + (1 - \beta_j)W_3) + (1 - \alpha_i)(\beta_j W_2 + (1 - \beta_j)W_4)] \end{aligned} \quad (13)$$

where $\lambda, \beta_j \in [0, 1]$. Equation (13) weighs two components. The first is the standard expected-utility calculation under fully strategic play. The second replaces citizen j 's mixed strategy, α_j , with citizen i 's initial belief that citizen j will cooperate, β_j . As common-knowledge rationality declines ($\lambda \rightarrow 0$), citizen i 's expected utility depends more on initial beliefs, β_j . Appendix B.7 provides the proofs.¹⁵

¹⁵The proofs show the critical thresholds that determine equilibrium selection. They trace equilibrium

Figure 2: Trace Plots



Note: Each panel traces the MSNE cutoff, α^* , as the weight on common-knowledge rationality, λ , varies. Blank regions indicate values outside the relevant stability set. The middle column collapses the collective-only and private-only monitoring cases into a single monitoring-benefit category for purposes of parsimony.

The stylized trace plots in Figure 2 assist with understanding how all the parameters interact with each other. For each combination of β_j and monitoring level, under full common-knowledge rationality (i.e., $\lambda = 1$) we can verify the presence of the three assurance game equilibria: $PSNE_{Cooperate}$, $PSNE_{Defect}$, and $MSNE$ (α^*). As common-knowledge rationality declines ($\lambda \rightarrow 0$), we trace the MSNE/indifference threshold (α^*) until 0 or 1, respectively, where it becomes a PSNE. Beyond that point, common-knowledge rationality (λ) is sufficiently small for citizen i to prefer one PSNE over another due to her beliefs about citizen j 's probability of cooperation, β_j .

Each row of Figure 2 illustrates how different initial beliefs about others' cooperation levels (β_j) affect coordination outcomes. Across all of its components 2a-2l, blank values for either $PSNE_{Cooperate}$ or $PSNE_{Defect}$ indicate the equilibrium is not sustained for all values of λ (conditional on β_j), thereby excluding it from the stability set.¹⁶ Under high values of β_j in Figures 2a-2c, the low-corruption equilibrium strategy ($PSNE_{Cooperate}$) makes up the stability set regardless of the level of monitoring. The opposite is true for low values of β_j . In Figures 2j-2l, introducing various forms of monitoring does not change the fact that only the high-corruption equilibrium is in the stability set. For these select cases where initial beliefs in others' cooperation are either exceedingly high or low, monitoring does not change the central outcome of determining which equilibrium makes up the stability set. Nonetheless, monitoring can substantively matter in those cases, too. To the extent that initial beliefs are sticky but responsive to policy, monitoring can facilitate an easier transition to a low-corruption equilibrium or avoid a retreat to a high-corruption equilibrium.

Meanwhile, the second and third rows of Figure 2 most markedly show the effect of monitoring. These rows depict sufficiently low values of β_j for the high-corruption equilibrium to be in the stability set when no monitoring is present (see Figures 2d and 2g).

correspondence as a function of λ and β_j , showcasing when unstable equilibria drop off.

¹⁶Medina (2007, 113) argues that an equilibrium is in the stability set when it provides a continuous feasible path for all values of λ . As common knowledge of rationality is phased in (rather than relaxed), players will need substantial impetus to depart from their original (optimal) strategy and opt for another optimal strategy. Although we find this logic to be most intuitive, we opted to gradually phase out for beliefs as full knowledge of rationality marks the starting point of our model (see Equation (10)).

However, with the addition of monitoring-derived collective and/or private benefits, the low-corruption equilibrium can replace its high-corruption counterpart in the stability set (see Figures 2e, 2f, and 2i).

Let us now show the visual logic from the trace plots probabilistically. As before, the MSNE (α_i^*) denotes the threshold separating the high- and low-corruption equilibria. If initial beliefs (β_j) fall below this threshold, they lie in the basin of attraction of the high-corruption equilibrium, which we can calculate as follows:

$$Pr(High\ Corruption) = F_{\beta_j}(\alpha_i^*) = \int_0^{\alpha_i^*} f(\beta_j) d\beta_j \quad (14)$$

Alternatively, if initial beliefs are higher than α_i^* , they lie in the basin of attraction of the low-corruption equilibrium, which we can calculate as follows:

$$Pr(Low\ Corruption) = 1 - F_{\beta_j}(\alpha_i^*) = \int_{\alpha_i^*}^1 f(\beta_j) d\beta_j \quad (15)$$

Conditional on β_j having a probability distribution bounded between 0 and 1, we can also calculate the change in probability of the low-corruption equilibrium occurring across different versions of the game. For instance, moving from the baseline game, Γ^A , to the one where citizens receive both collective and private monitoring benefits, Γ^D , we obtain:

$$\begin{aligned} & Pr(Low\ Corruption\ \Gamma^D) - Pr(Low\ Corruption\ \Gamma^A) \\ &= (1 - F_{\beta_j}(\alpha_i^{*,D})) - (1 - F_{\beta_j}(\alpha_i^{*,A})) \\ &= \int_{\alpha_i^{*,D}}^1 f(\beta_j) d\beta_j - \int_{\alpha_i^{*,A}}^1 f(\beta_j) d\beta_j \geq 0 \end{aligned} \quad (16)$$

Because monitoring lowers the mixed-strategy cutoff, it expands the set of initial beliefs for which the low-corruption equilibrium is the best response. Technically, per Appendix B.6, p shifts the threshold more than an equal increase in c . Nonetheless, for a given distribution F_{β_j} , both types of monitoring increase the probability of the low-corruption equilibrium

relative to the baseline game under most circumstances.¹⁷

6. Discussion

The corruption literature’s recent recalibration away from top-down, principal-agent approaches has yielded significant insights but also some challenges. The shift toward equilibrium-oriented, collective-action approaches has improved understanding of the basic nature of corruption (e.g., [Mungiu-Pippidi, 2013](#); [Fisman and Golden, 2017](#); [Persson, Rothstein and Teorell, 2019](#)). At the same time, they tend to locate the relevant equilibrium at an overly high level of aggregation. By emphasizing society-wide norms and “big-bang” reforms, existing collective-action accounts better explain persistence than incremental change and outcome heterogeneity. In this respect, they face a level-of-analysis problem similar to the one that afflicts perception-based corruption indicators (see [Gingerich, 2013a](#)).

Our model addresses the level-of-analysis challenge by shifting attention from society-level equilibria to the citizen-level payoffs and beliefs that shape equilibrium selection and stability. In doing so, we retain the multiple-equilibria, assurance game structure that the collective-action literature emphasizes. A main finding is that monitoring can alter the threshold separating high- and low-corruption equilibria, all without eliminating the high-corruption equilibrium. The finding helps explain why incremental interventions can matter even in the presence of systemic corruption. Big-bang reforms can sometimes be valuable but are not the only route to progress (see [Appendix A](#)). Reforms can also work by changing payoffs in particular agencies, projects, procurement processes, service-delivery settings, or bureaucratic encounters ([Bersch, 2016](#); [Taylor, 2018](#)).

We derive another key policy insight by unbundling monitoring benefits by their incidence. Building on [Ostrom \(1990\)](#) and [Rothstein’s \(2021\)](#) reappropriation of common-pool-

¹⁷Note that the difference of cumulative distribution functions of β_j is only zero when the distribution of β_j does not have any mass within the range of $\beta \in (\alpha_i^{*,D}, \alpha_i^{*,A}]$. For $\beta_j \sim \text{Beta}(\alpha, \beta)$, where α and β are the distribution’s shape parameters, and not parameters in our model, this is a possible outcome. Conversely, if $\beta_j \sim \text{U}(0, 1)$, all values for β_j between 0 and 1 have non-zero probability, in which case Equation (16) is strictly positive. We address [Bauhr and Grimes’s \(2014\)](#) concern about transparency in the next section.

resource solutions for corruption control, monitoring can generate both collective and private benefits. Collective benefits (c) are non-excludable and subject to free-riding. Private benefits (p) are excludable and only attach to individuals taking costly action against corruption. The distinction matters because c and p enter payoff structures differently. Holding the perceived size of the benefit constant, our model shows that p lowers the cooperation threshold more than c , as c leaks to free-riders while p accrues only to cooperators.

The c versus p benefits distinction, along with its incidence asymmetry, provides a heuristic for policy design. For example, if c and p are both equally close to zero in a highly-corrupt setting, then perhaps it is more useful to invest in NGOs that provide legal support to corruption victims and other mechanisms that enhance p . That may be especially the case under situations where transparency about corrupt processes demobilizes citizens (Bauhr and Grimes, 2014). Alternatively, if the binding constraint is weak process-level accountability, then perhaps investing in audits, e-procurement systems, or other measures that enhance c are germane. Ideally, most anti-corruption reforms will invest in both c and p , as they are not substitutes. Even in highly corrupt settings, citizens do not always accede to bribe requests (Erllich, Gans-Morse and Nichter, 2025), so the question is not whether to act but how. Frequently, effective reforms may need to be “second-best”: perhaps not theoretically ideal, but targeted at the relevant constraint, taking into account power dynamics (Rodrik, 2007; Levy, 2014; Dercon, 2022).

7. Conclusion

What reduces corruption? A predominant answer and policy response is top-down monitoring in line with the principal-agent model, but a now well-established collective-action literature rightfully asks: what happens when the principals are unprincipled?

We devise a multiple-equilibria assurance model that integrates collective-action, principal-agent, and some functionalist approaches to corruption. Our model internalizes the ‘unprincipled principals’ critique while addressing the tendency of collective-action approaches to

locate the relevant equilibria at overly high levels of aggregation. Our solution to this level-of-analysis problem is two-fold. First, we distinguish between citizen-level monitoring benefits that are collective and private. Both can lower the threshold for taking costly action against corruption, but our model shows that equally-sized private benefits are more effective per unit because they do not spill over to free-riders. Second, our model moves beyond merely identifying equilibria to examining their selection and stability. The result is a model that distinguishes between the incidence of monitoring benefits and explains how monitoring and other incremental interventions can reduce corruption without requiring a ‘big-bang’ reform.

Going forward, future research can build not just on the distinction between collective and private monitoring benefits, but also on their empirical underpinnings. If relevant interventions do not alter citizens’ perceived payoffs or beliefs about whether costly action against corruption is worthwhile, then another intervention may be more appropriate. In deciding among these interventions, the literatures on problem-driven iterative adaptation and political settlements provide useful guidance: reforms should be local, constraint-sensitive, and attentive to elite interests ([Andrews, Pritchett and Woolcock, 2017](#); [Khan, 2018](#)). While corruption can be systemic, there is often within-country heterogeneity to exploit for progress ([Levy, 2014](#); [Bersch, 2016](#)).

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Appendices

Appendix A Illustrative Pathways to Lower-Corruption Outcomes App-2

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Appendix A Illustrative Pathways to Lower-Corruption Outcomes

Table A.1: Country-Level Paths to Lower-Corruption Outcomes

Country	Critical Period(s)	How the Country Shifted to a Lower-Corruption Path	Maintained?
Denmark	1658-1665, 1814, 1849	Loss in wars against Sweden; top-down reform initiated by kings; drafting of a new constitution following demonstrations	Yes
Sweden	1810-1850	Losing the 1808-1809 war against Russia, followed by a series of reforms	Yes
Great Britain	1780-1883	Civil service reform; legislation; a secret ballot; suffrage reform, resulting in the decline of clientelism and more funds for public services	Yes
France	1791-1975	The French Revolution; gradual decline of patronage appointments; construction of impartial institutions	Yes
Germany	1919, 1945	End of World War I and World War II.	Yes
Spain	1975-1978	End of the dictatorship of Francisco Franco.	Yes
Portugal	1974	End of the Estado Novo dictatorship, fueled by the Carnation Revolution	Yes
Ireland	1995-2007, 2012	Privatization, market reforms, corruption law reforms introduced (2012), the rise of the Celtic Tiger ended up increasing corruption, Council of Europe's Greco initiative (anti-corruption program)	Yes
Italy	1992-1996	The Clean Hands scandal, prompted by the arrest of one well-connected individual, who provided information that led to the arrest of hundreds and changed the party system	No
Estonia	1990-1995	Tax reform; e-governance; procurement reform; privatization	Yes

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Table A.1: Country-Level Paths to Lower-Corruption Outcomes – *continued*

Country	Critical Period(s)	How the Country Shifted to a Lower-Corruption Path	Maintained?
Georgia	2004-2008	The Georgian Transition, followed by a “big bang” approach from President Mikheil Saakashvili (i.e., large-scale dismissal of civil servants, televised arrests, and e-governance)	Mixed, with creeping authoritarianism and human rights issues.
Tunisia	2011-2014	Citizen demonstrations over autocratic rule fueled the Arab Spring	Mixed; some patronage remains a challenge
Botswana	1966-present	Excellent natural resource management; protection of property rights; transparent policy-making; management of potential ethnic tensions. Strong and independent judiciary, and an exceptional rule of law.	Regular scandals imperil progress. The discovery of rich resource deposits has also led to an increase in corruption.
United States	1870-1920	The regulation of patronage appointments through the Pendleton Act; the press; the Progressivist movement; successful prosecutions.	Yes, though the role of money in politics is significant
Hong Kong	1974-1977	Egregious malfeasance by the head of police, which prompted the creation of an independent anti-corruption agency and many subsequent arrests	Yes
Taiwan	1992-	Civil service reform; high-level corruption initiatives; legislation; party system change	Yes
Singapore	1959-1990	Authoritarian leader Lee Kuan Yew pushed through a series of reforms	Yes
South Korea	1961-2003	Education; import-substitution industrialization that fueled economic growth; market reforms; legislation; protests	Yes
Japan	1945-1993,	Loss of World War 2; MacArthur Plan; resolution of a series of corruption scandals involving kickbacks and vote-buying.	Yes

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Table A.1: Country-Level Paths to Lower-Corruption Outcomes – *continued*

Country	Critical Period(s)	How the Country Shifted to a Lower-Corruption Path	Maintained?
Chile	1984-1990	Economic liberalization; privatization; loss of natural resource rents; democratic and authoritarian legacies from previous periods	Yes
Uruguay	1984	Fiscal/tax system consolidation; privatization; a democratic history; an educated and active citizenry; loss of patronage funds.	Yes
United Arab Emirates	1989-2016	E-Governance; institutional development; avoided the misuse of oil rents by investing in physical capital and institutional fabric, establishment of the Prosecution of Public Funds Office to prosecute corruption, adopting a federal penal code anchored on UN Convention Against Corruption (UNCAC)	Yes
Namibia	2000-2010	Effective legislation (Anti-Corruption Act, 2003, under the ambit of the Constitution, Electoral Act, 2014) and enforcement bodies (e.g. the Anti-Corruption Commission)	Yes
Cape Verde	1990-present	A wave of democratization, introduction of parties, e-governance	Yes
Costa Rica	1948-present	civil war of 1948, followed by new regime and electoral institute to deter election fraud; improvement in professionalism of judicial branch staff; audit and constitutional improvements following scandals; signing of OECD anti-bribery measures	Mostly

Sources: McCarthy (2003), Theriault (2003), Lizzeri and Persico (2004), Glaeser and Goldin (2007), Baker (2009), Rothstein (2011b), Acemoglu and Robinson (2012), Grindle (2012), Weyland (2012), Camp, Dixit and Stokes (2014), Al Athmay (2015), Arroyo Chacón (2015), Mungiu-Pippidi (2015, 2016), Polus, Kopinski and Tycholiz (2015), Rothstein and Teorell (2015), Soto and Haouas (2016), Teorell and Rothstein (2015), Soto and Haouas (2016), Fisman and Golden (2017), Fukuyama (2018), Masoud (2018), Nyblade and Reed (2008), Wilson and Villarreal (2017)

Appendix B Mathematical Proofs for Theoretical Model

B.1 Basic Collective Action Game: Version Γ^A

		Citizen 2	
		<i>Cooperate/</i>	<i>Defect/</i>
		<i>Take Action</i>	<i>Do Nothing</i>
Citizen 1	<i>Cooperate/Take Action</i>	$t - r,$ $t - r$	$s - r,$ $g - b + s$
	<i>Defect/Do Nothing</i>	$g - b + s,$ $s - r$	$g - b,$ $g - b$

Where we define:

$$\begin{aligned}
 W_1 &= t - r \\
 W_2 &= g - b + s \\
 W_3 &= s - r \\
 W_4 &= g - b;
 \end{aligned}
 \tag{B.1}$$

We further define all parameters to be non-negative:

$$\{t, r, g, b, s\} \in (0, \infty) : \tag{B.2}$$

And, in order to maintain the characteristics of an assurance game, as well as meet the

assumptions of collective action games (Medina, 2007, 53)¹⁸, we assume:

$$\begin{aligned}
 W_1 &> W_4 > W_3, \\
 W_2 &> W_4 > W_3, \\
 W_1 &> W_2
 \end{aligned}
 \tag{B.3}$$

B.1.1 Pure Strategy Nash Equilibria for Version Γ^A

$\oplus (C, C)$ is a PSNE if and only if for each player $i (i \in \{1, 2\})$:

$$\begin{aligned}
 u_i(C_i|C_j) &> u_i(D_i|C_j) \\
 &= t - r > g - b + s, \\
 &= t > s + r \text{ (centering game at zero (g-b=0))}
 \end{aligned}
 \tag{B.4}$$

$\oplus (D, D)$ is a PSNE, if and only if for each player $i (i \in \{1, 2\})$:

$$\begin{aligned}
 u_i(D_i|D_j) &> u_i(C_i|D_j) \\
 &= g - b > s - r, \\
 &= 0 > s - r \text{ (centering game at zero (g-b=0))} \\
 &= r > s
 \end{aligned}
 \tag{B.5}$$

B.1.2 Mixed Strategy Nash Equilibrium for Version Γ^A

Player i chooses to play C [i.e. report the bureaucrat/refuse to pay the bribe] with probability α_i , such that player j ($i \neq j$) is indifferent between playing C [reporting/refusing to pay the bribe] and playing D [paying the bribe]:

¹⁸Medina (2007) points out that for there to be any hope for cooperation, at least some members of society will need to prefer mutual cooperation over unilateral defection (i.e., $W_{1i} > W_{2i}$). To indicate this *individual* preference, he uses the subscript i . Since our focus here lies on the feasibility of generating greater propensity to cooperate, rather than teasing out thresholds for sufficient cooperation, we have dropped the subscript for ease of exposition.

$$\begin{aligned}
EU_j(C) &= EU_j(D) \\
\alpha_i(W_1) + (1 - \alpha_i)(W_3) &= \alpha_i(W_2) + (1 - \alpha_i)(W_4) \\
\alpha_i(t - r) + (1 - \alpha_i)(s - r) &= \alpha_i(g - b + s) + (1 - \alpha_i)(g - b) \\
\alpha_i(t - s) + (s - r) &= \alpha_i(s) + (g - b) \\
\alpha_i^{*,A} &= \frac{g - b + r - s}{t - 2s}
\end{aligned} \tag{B.6}$$

Which is a valid mixing probability under the previously made assumptions when players are fully rational. Specifically, centering the game at zero again, $\alpha_i^{*,A} = \frac{r-s}{t-2s}$, where $r - s > 0$ as $r > s$ by assumption (see above); and $t - 2s > 0$, since $t > s + r$, and $r > s$.

In such a case, their best response is to choose α_i as follows:

$$\alpha_i^*(\alpha_j) = \begin{cases} 1 & \text{if } \alpha_j > \frac{g-b+r-s}{t-2s} \\ [0, 1] & \text{if } \alpha_j = \frac{g-b+r-s}{t-2s} \\ 0 & \text{if } \alpha_j < \frac{g-b+r-s}{t-2s} \end{cases} \tag{B.7}$$

B.2 Partially Augmented Collective Action Game: Version Γ^B

		Citizen 2	
		<i>Cooperate/</i>	<i>Defect/</i>
		<i>Take Action</i>	<i>Do Nothing</i>
Citizen 1	<i>Cooperate/Take Action</i>	$t - r + c,$ $t - r + c$	$s - r + c,$ $g - b + s + c$
	<i>Defect/Do Nothing</i>	$g - b + s + c,$ $s - r + c$	$g - b,$ $g - b$

Where:

$$\begin{aligned}
 W_1 &= t - r + c \\
 W_2 &= g - b + s + c \\
 W_3 &= s - r + c \\
 W_4 &= g - b;
 \end{aligned} \tag{B.8}$$

Again, we further define all parameters to be non-negative:

$$\{t, r, g, b, s, c\} \in (0, \infty) : \tag{B.9}$$

And maintain the assumptions made previously so as to sustain the assurance game, and collective action game, requirements (Eq. B.3).

B.2.1 Pure Strategy Nash Equilibria for Version Γ^B

$\oplus (C, C)$ is a PSNE if and only if for each player $i (i \in \{1, 2\})$:

$$\begin{aligned}
 u_i(C_i|C_j) &> u_i(D_i|C_j) \\
 &= t - r + c > g - b + s + c, \\
 &= t > s + r \text{ (centering game at zero (g-b=0))}
 \end{aligned} \tag{B.10}$$

$\oplus (D, D)$ is a PSNE, if and only if for each player $i (i \in \{1, 2\})$:

$$\begin{aligned}
 u_i(D_i|D_j) &> u_i(C_i|D_j) \\
 &= g - b > s - r + c, \\
 &= 0 > s - r + c \text{ (centering game at zero (g-b=0))} \\
 &= r > s + c
 \end{aligned} \tag{B.11}$$

B.2.2 Mixed Strategy Nash Equilibrium for Version Γ^B

Player i chooses to play C [i.e. report the bureaucrat/refuse to pay the bribe] with probability α_i , such that player j ($i \neq j$) is indifferent between playing C [reporting/refusing to pay the bribe] and playing D [paying the bribe]:

$$\begin{aligned}
 EU_j(C) &= EU_j(D) \\
 \alpha_i(W_1) + (1 - \alpha_i)(W_3) &= \alpha_i(W_2) + (1 - \alpha_i)(W_4) \\
 \alpha_i(t - r + c) + (1 - \alpha_i)(s - r + c) &= \alpha_i(g - b + s + c) + (1 - \alpha_i)(g - b) \quad (\text{B.12}) \\
 \alpha_i(t - s) + (s - r + c) &= \alpha_i(s + c) + (g - b) \\
 \alpha_i^{*,B} &= \frac{g - b + r - s - c}{t - 2s - c}
 \end{aligned}$$

Which is a valid mixing probability under the previously made assumptions as long as $t - 2s - c > 0$. Specifically, centering the game at zero again, $\alpha_i^{*,B} = \frac{r - s - c}{t - 2s - c}$, where $r - s - c > 0$ and $r > s + c$ by assumption (see above).

Now, their best response under full common-knowledge is to choose α_i as follows:

$$\alpha_i^*(\alpha_j) = \begin{cases} 1 & \text{if } \alpha_j > \frac{g - b + r - s - c}{t - 2s - c} \\ [0, 1] & \text{if } \alpha_j = \frac{g - b + r - s - c}{t - 2s - c} \\ 0 & \text{if } \alpha_j < \frac{g - b + r - s - c}{t - 2s - c} \end{cases} \quad (\text{B.13})$$

B.3 Partially Augmented Collective Action Game: Version Γ^C

		Citizen 2	
		<i>Cooperate/</i>	<i>Defect/</i>
		<i>Take Action</i>	<i>Do Nothing</i>
Citizen 1	<i>Cooperate/Take Action</i>	$t - r + p,$ $t - r + p$	$s - r + p,$ $g - b + s$
	<i>Defect/Do Nothing</i>	$g - b + s,$ $s - r + p$	$g - b,$ $g - b$

Where:

$$\begin{aligned}
 W_1 &= t - r + p \\
 W_2 &= g - b + s \\
 W_3 &= s - r + p \\
 W_4 &= g - b;
 \end{aligned}
 \tag{B.14}$$

Still, we further define all parameters to be non-negative:

$$\{t, r, g, b, s, p\} \in (0, \infty) : \tag{B.15}$$

And maintain the assumptions made previously (Eq. B.3) keeping with the necessary collective action and assurance game assumptions.

B.3.1 Pure Strategy Nash Equilibria for Version Γ^C

$\oplus (C, C)$ is a PSNE if and only if for each player $i (i \in \{1, 2\})$:

$$\begin{aligned}
 u_i(C_i|C_j) &> u_i(D_i|C_j) \\
 &= t - r + p > g - b + s, \\
 &= t > s + r - p \text{ (centering game at zero (g-b=0))}
 \end{aligned} \tag{B.16}$$

$\oplus (D, D)$ is a PSNE, if and only if for each player $i (i \in \{1, 2\})$:

$$\begin{aligned}
 u_i(D_i|D_j) &> u_i(C_i|D_j) \\
 &= g - b > s - r + p, \\
 &= 0 > s - r + p \text{ (centering game at zero (g-b=0))} \\
 &= r > s + p
 \end{aligned} \tag{B.17}$$

B.3.2 Mixed Strategy Nash Equilibrium for Version Γ^C

Player i chooses to play C [i.e. report the bureaucrat/refuse to pay the bribe] with probability α_i , such that player j ($i \neq j$) is indifferent between playing C [reporting/refusing to pay the bribe] and playing D [paying the bribe]:

$$\begin{aligned}
 EU_j(C) &= EU_j(D) \\
 \alpha_i(W_1) + (1 - \alpha_i)(W_3) &= \alpha_i(W_2) + (1 - \alpha_i)(W_4) \\
 \alpha_i(t - r + p) + (1 - \alpha_i)(s - r + p) &= \alpha_i(g - b + s) + (1 - \alpha_i)(g - b) \\
 \alpha_i(t - s) + (s - r + p) &= \alpha_i(s) + (g - b) \\
 \alpha_i^{*,C} &= \frac{g - b + r - s - p}{t - 2s}
 \end{aligned} \tag{B.18}$$

Which is a valid mixing probability under the previously made assumptions when players are fully rational. Specifically, centering the game at zero again, $\alpha_i^{*,C} = \frac{r-s-p}{t-2s}$, where

$r - s - p > 0$ as $r > s + p$ by assumption (see above); and $t - 2s > 0$, since $t > s + r$, and $r > s$.

Now, their best response under full common-knowledge rationality is to choose α_i as follows:

$$\alpha_i^*(\alpha_j) = \begin{cases} 1 & \text{if } \alpha_j > \frac{g-b+r-s-p}{t-2s} \\ [0, 1] & \text{if } \alpha_j = \frac{g-b+r-s-p}{t-2s} \\ 0 & \text{if } \alpha_j < \frac{g-b+r-s-p}{t-2s} \end{cases} \quad (\text{B.19})$$

B.4 Fully Augmented Collective Action Game: Version Γ^D

		Citizen 2	
		<i>Cooperate/</i>	<i>Defect/</i>
		<i>Take Action</i>	<i>Do Nothing</i>
Citizen 1	<i>Cooperate/Take Action</i>	$t - r + p + c,$ $t - r + p + c$	$s - r + p + c,$ $g - b + s + c$
	<i>Defect/Do Nothing</i>	$g - b + s + c,$ $s - r + p + c$	$g - b,$ $g - b$

Where:

$$\begin{aligned} W_1 &= t - r + p + c \\ W_2 &= g - b + s + c \\ W_3 &= s - r + p + c \\ W_4 &= g - b \end{aligned} \quad (\text{B.20})$$

Here, too, we further define all parameters to be non-negative:

$$\{t, r, g, b, s, p, c\} \in (0, \infty) : \quad (\text{B.21})$$

And maintain the assumptions made previously (Eq. B.3).

B.4.1 Pure Strategy Nash Equilibria for Version Γ^D

$\oplus (C, C)$ is a PSNE, because for each player $i (i \in \{1, 2\})$:

$$\begin{aligned}
 u_i(C_i|C_j) &> u_i(D_i|C_j) \\
 &= t - r + p + c > g - b + s + c \\
 &= t - r + p + c > s + c \text{ (applying centering at } g-b = 0) \\
 &= t > s + r - p \text{ (True by base game's assumption, as } t > s + r)
 \end{aligned} \tag{B.22}$$

$\oplus (D, D)$ is a PSNE, if and only if for each player $i (i \in \{1, 2\})$:

$$\begin{aligned}
 u_i(D_i|D_j) &> u_i(C_i|D_j) \\
 &= g - b > s - r + p + c \\
 &= 0 > s - r + p + c \text{ (applying centering at } g-b = 0) \\
 &= r - p - c > s
 \end{aligned} \tag{B.23}$$

B.4.2 Mixed Strategy Nash Equilibrium for Version Γ^D

Again, player i chooses to play C with probability α_i such that player j is indifferent between playing C and D :

$$\begin{aligned}
 EU_j(C) &= EU_j(D) \\
 \alpha_i(W_1) + (1 - \alpha_i)(W_3) &= \alpha_i(W_2) + (1 - \alpha_i)(W_4) \\
 \alpha_i(t - r + p + c) + (1 - \alpha_i)(s - r + p + c) &= \alpha_i(g - b + s + c) + (1 - \alpha_i)(g - b) \\
 \alpha_i(t - s) + (s - r + p + c) &= \alpha_i(s + c) + g - b \\
 \alpha_i(t - 2s - c) &= g - b + r - s - p - c \\
 \alpha_i^{*,D} &= \frac{g - b + r - s - p - c}{t - 2s - c}
 \end{aligned} \tag{B.24}$$

Which is a valid mixing probability as long as $t - 2s - c > 0$, and (as previously assumed) $r > s + p + c$. Again, under full common-knowledge rationality player i 's best response is:

$$\alpha_i^*(\alpha_j) = \begin{cases} 1 & \text{if } \alpha_j > \frac{g-b+r-s-p-c}{t-2s-c} \\ [0, 1] & \text{if } \alpha_j = \frac{g-b+r-s-p-c}{t-2s-c} \\ 0 & \text{if } \alpha_j < \frac{g-b+r-s-p-c}{t-2s-c} \end{cases} \quad (\text{B.25})$$

B.5 Relaxing the Assumption of Common-Knowledge Rationality

Generally, if players are not *fully* rational, player i 's utility functions can be re-written as a weighted average of her payoffs under rationality, and her payoffs conditional upon her beliefs about the likelihood of cooperation by the other player (j). Further, let $\lambda \in [0, 1]$ denote the degree to which actors are fully rational:

$$\begin{aligned} u_i^\lambda(\alpha_i, \alpha_j, \beta_j) &= \lambda[\alpha_i(\alpha_j(W_1) + (1 - \alpha_j)(W_3)) \\ &\quad + (1 - \alpha_i)(\alpha_j(W_2) + (1 - \alpha_j)(W_4))] \\ &\quad + (1 - \lambda)[\alpha_i(\beta_j(W_1) + (1 - \beta_j)(W_3)) \\ &\quad + (1 - \alpha_i)(\beta_j(W_2) + (1 - \beta_j)(W_4))] \end{aligned} \quad (\text{B.26})$$

First, note that when $\lambda = 1$, Equation (B.26) simplifies to the optimal mixing probabilities for each player noted in the MSNE of a given game. Meanwhile, let $\lambda = 0$ so as to generate player i 's utility function when strategies entirely depend on beliefs about the other player's behavior rather than rationality:

$$\begin{aligned} u_i^{\lambda=0}(\alpha_i, \beta_j) &= \alpha_i(\beta_j(W_1) + (1 - \beta_j)(W_3)) + (1 - \alpha_i)(\beta_j(W_2) + (1 - \beta_j)(W_4)) \\ &= \alpha_i\beta_jW_1 - \alpha_i\beta_jW_3 - \alpha_i\beta_jW_2 + \alpha_i\beta_jW_4 + \alpha_iW_3 - \alpha_iW_4 + \beta_jW_2 - \beta_jW_4 + W_4 \end{aligned} \quad (\text{B.27})$$

B.6 Collective vs. Private Monitoring Benefits

In order to assess whether the provision of private or collective benefits is more likely to reduce the likelihood of corruption, we must ascertain the conditions under which the Mixed Strategy Nash Equilibria of Equations (4) and (5) exceed one another:¹⁹

$$\begin{aligned}
\alpha_i^{*'} &> \alpha_i^{*''} \\
\frac{r-s-c}{t-2s-c} &> \frac{r-s-p}{t-2s} \\
\frac{(r-s-c)(t-2s)}{(t-2s-c)(t-2s)} &> \frac{(r-s-p)(t-2s-c)}{(t-2s)(t-2s-c)} \\
\frac{rt-st-ct-2rs+2s^2+2cs}{(t-2s-c)(t-2s)} &> \frac{rt-2rs-rc-st+2s^2+cs-pt-2ps+pc}{(t-2s-c)(t-2s)}
\end{aligned} \tag{B.28}$$

Upon simplifying and dropping the denominator—which is always positive, as assumed above so as to maintain the assurance game format of the game—we obtain:

$$\begin{aligned}
cs - ct - cr - pc + pt - 2ps &> 0 \\
p(t-2s-c) &> c(t-s-r) \\
p &> \frac{c(t-s-r)}{t-2s-c}
\end{aligned} \tag{B.29}$$

What we can discern from Equation (B.29) is that when private benefits (p) are sufficiently large relative to collective benefits (c), providing private benefits is more likely to be *effective* in reducing corruption than collective benefits. Further, we can show that the multiplier on c , $\frac{t-s-r}{t-2s-c}$, is always smaller than unity:

$$\begin{aligned}
t-s-r &< t-2s-c \\
s+c &< r
\end{aligned} \tag{B.30}$$

¹⁹Note that the greater the reduction in the mixing probability, the more likely a policy is to reduce corruption as we relax the rationality assumption. See proofs in Appendices B.2.2 and B.3.2.

This inequality follows from Equation B.11 and always holds due to the assumptions necessary to uphold the basic form of the augmented assurance game, Version Γ^B .

In summary, private benefits to individuals, such as whistleblower protections, can diminish the likelihood that citizens will acquiesce to bribe requests by corrupt bureaucrats. Similarly, collective benefits, such as increased bureaucrat monitoring, can also reduce this likelihood. When citizens value collective and private benefits equally, providing private benefits is relatively more likely to be effective than providing collective benefits. Collective and private benefits, however, are not substitutes. Even as the basic form of the assurance game is maintained, the effects of both collective and private benefits are additive in combating corruption.

B.7 Critical Belief Thresholds

Under the definition of stability sets, an equilibrium is considered stable if and only if it presents an optimal strategy selection that maximizes utility for all levels of beliefs over player j 's common-knowledge rationality (λ). In Medina's (2007, 111) words, "a profile of initial belief conditions β is mapped into an equilibrium α^* of a game if there is a continuous path leading from the equilibrium of the game when $\lambda = 0$ to α^* when $\lambda = 1$." As such, knowing that our game has three equilibria when $\lambda = 1$, if we can identify whichever equilibria maximize utility when $\lambda = 0$, we know which equilibria falls within the stability set.

To show this, we start with Equation (B.27), where $\lambda = 0$ and maximize it with respect to α_i :

$$\begin{aligned} \frac{\delta u_i}{\delta \alpha_i} &= \beta_j(W_1) - \beta_j(W_3) - \beta_j(W_2) + \beta_j(W_4) + W_3 - W_4 \\ &= \beta_j(W_1 - W_2 + W_4 - W_3) + W_3 - W_4 \end{aligned} \tag{B.31}$$

Next, for player i to prefer cooperation when $\lambda = 0$ (and only beliefs about others' willingness to cooperate inform player i 's strategy), we know that the derivative in Equa-

tion (B.31) above must be positive. Solving for β_j , we can identify the game's threshold where this is true:

$$\begin{aligned} \frac{\delta u_i}{\delta \alpha_i} &= \beta_j(W_1 - W_2 + W_4 - W_3) + W_3 - W_4 > 0 \\ \beta_j &> \frac{W_4 - W_3}{W_1 - W_2 + W_4 - W_3} = \alpha_i^* \end{aligned} \tag{B.32}$$

Recall that per Equation (B.3), the assumptions made to maintain assurance game form are $W_1 > W_2 > W_4 > W_3$, so we can derive that both the denominator and numerator will always be positive, and that $0 < \beta < 1$, which is required for it to be a valid belief (i.e., a probability).

Conversely, the reverse is true and player i prefers defection if the derivative in Equation (B.31) is negative. Further note that this threshold was also defined as the game's MSNE (α_i^*) under complete common rationality (i.e., $\lambda = 1$) in Equation (3).²⁰ This threshold is the point where beliefs β_j either map into the high- or low-corruption equilibrium. Recalling Medina's (2007) language, any initial belief conditions that are mapped into a given corruption equilibrium will be stable. Note also that when $\beta_j = \alpha_i$, gradually introducing beliefs doesn't change player i 's calculus, as one weighted portion of the utility function is replaced by a (numerically) identical utility function.²¹

Additionally, we can compute the shape of the path for the MSNE as beliefs are gradually introduced as well as the point for a given game where the MSNE disappears completely, and only the stable equilibrium remains. First, to compute the exact path of the MSNE, we

²⁰Naturally, this is by construction and not coincidence, since beliefs over others' willingness to cooperate (β_j) have entirely replaced player j 's rationality-based strategy intent on making player i indifferent between cooperation and defection (α_j) from the complete common rationality utility function (see Equation (10)) where $\lambda = 1$.

²¹Moreover, because β_j is a continuous random variable, its distribution has no point masses, such that β_j being exactly one value has zero probability of occurring.

must return to player i 's generalized utility function (Equation (B.26)):

$$\begin{aligned}
u_i^\lambda(\alpha_i, \alpha_j, \beta_j) = & \lambda[\alpha_i(\alpha_j(W_1) + (1 - \alpha_j)(W_3)) \\
& + (1 - \alpha_i)(\alpha_j(W_2) + (1 - \alpha_j)(W_4))] \\
& + (1 - \lambda)[\alpha_i(\beta_j(W_1) + (1 - \beta_j)(W_3)) \\
& + (1 - \alpha_i)(\beta_j(W_2) + (1 - \beta_j)(W_4))]
\end{aligned} \tag{B.33}$$

Upon simplifying and computing the partial derivative with respect to α_i to calculate the optimal strategy for player i , we get:

$$\begin{aligned}
\frac{\delta u_i}{\delta \alpha_i} = & \alpha_j \lambda (W_1 - W_2 + W_4 - W_3) \\
& - \beta_j \lambda (W_1 - W_2 + W_4 - W_3) \\
& + \beta_j (W_1 - W_2 + W_4 - W_3) \\
& + W_3 - W_4
\end{aligned} \tag{B.34}$$

Upon setting this partial derivative to zero, and isolating the parameters from the payoffs, we get:

$$\begin{aligned}
\frac{\delta u_i}{\delta \alpha_i} = & 0 \\
\lambda(\alpha_j - \beta_j) + \beta_j = & \frac{W_4 - W_3}{W_1 - W_2 + W_4 - W_3}
\end{aligned} \tag{B.35}$$

Now, for brevity, let $z = \frac{W_4 - W_3}{W_1 - W_2 + W_4 - W_3}$. Solving for α_j , we get:

$$\alpha_j = \frac{1}{\lambda}(z - \beta_j) + \beta_j \tag{B.36}$$

As we can see, under full common knowledge rationality ($\lambda = 1$), this equation simplifies to the MSNE (α^* or z).²² For any other values of λ , a curve begins to take shape, as

²²Note also that the trace equation is undefined for $\lambda = 0$, but this is unproblematic: when $\lambda = 0$, player i 's

documented for some example cases in Figure 2. Each of these curves ultimately either reach $\alpha = 1$, the cooperative PSNE, or $\alpha = 0$, the defecting one. Next, we can compute for what values of λ the MSNE is defined.

To do this, we need to limit α_j (or α_i , due to symmetry) to values that are valid mixing probabilities, i.e., $\alpha_j \in (0, 1)$. We begin by looking at the case of $z > \beta_j$, where we expect the low-corruption equilibrium to disappear as $\lambda \rightarrow 0$.

First, we can start by probing the lower limit of $\alpha_j \geq 0$:

$$\alpha_j = \frac{1}{\lambda}(z - \beta_j) + \beta_j \geq 0 \quad (\text{B.37})$$

Knowing that β_j and λ are defined as strictly positive, and also knowing that in this case $z > \beta_j$, we know that this equation is always positive does not impose a limit on α_j . That is, when $z > \beta$, for all values of λ , $\alpha_i = 0$ is an equilibrium. Next, we look at the upper limit of $\alpha_j \leq 1$:

$$\alpha_j = \frac{1}{\lambda}(z - \beta_j) + \beta_j \leq 1 \quad (\text{B.38})$$

Upon isolating λ , we get:

$$\lambda \geq \frac{z - \beta_j}{1 - \beta_j} \quad (\text{B.39})$$

Recalling that $\beta \in (0, 1)$, and that $z = \frac{W_4 - W_3}{W_1 - W_2 + W_4 - W_3}$, we know that $z \in (0, 1)$. In addition, since $z > \beta_j$ for the case we are studying here, we know that Equation (B.38), imposes a limit on the values that λ can take. Namely, when $\lambda < \frac{z - \beta_j}{1 - \beta_j}$, the high-corruption PSNE ($\alpha = 0$) and MSNE (α^*) are no longer preferred, and thus not stable over the full range of λ .

utility function depends on β_j , not on player j 's optimizing strategy α_j .

Meanwhile, as we showed earlier in this section, we know that when $z < \beta_j$, it is the low-corruption that lacks stability over the range of λ . To illustrate this, we can again start with the lower limit on α_j :

$$\alpha_j = \frac{1}{\lambda}(z - \beta_j) + \beta_j \geq 0 \quad (\text{B.40})$$

Upon isolating λ , this time, we get:

$$\lambda \geq 1 - \frac{z}{\beta_j} \quad (\text{B.41})$$

Since $(z, \beta_j) \in (0, 1)$, and $z < \beta_j$ (for this exercise), we know that Equation (B.41) imposes a minimal level of λ such that α_j remains a valid mixing probability. That is, when $\lambda < 1 - \frac{z}{\beta_j}$, the low-corruption equilibrium and MSNE are no longer preferred, and thus not stable over the entire range of λ . Meanwhile, probing for potential upper range limits on α_j , we can look at our previous finding in Equation (B.39):

$$\lambda \geq \frac{z - \beta_j}{1 - \beta_j} \quad (\text{B.42})$$

Here, however, since $z, \beta_j \in (0, 1)$ and $z < \beta_j$, the right-hand term becomes negative, thus never imposing a limit on λ .

In summary, this section has shown that there is a critical threshold, the MSNE, which determines whether initial belief conditions map onto the low-corruption or high-corruption equilibrium. This matters, because upon gradually introducing beliefs into player i 's utility function, these initial belief conditions determine which equilibrium remains stable. In this section, we further showed how MSNEs can be computed as these beliefs are gradually introduced (i.e., $\lambda \rightarrow 0$). Finally, we illustrated at what levels of belief introduction the MSNE and the (unstable) PSNE disappear. These findings should be considered together

with our findings on the power of private and collective benefits introduced by monitoring to edge the MSNE closer towards zero, and thus fostering more cooperative anti-corruption behavior.

B.8 Calculating Stability Sets: A Numerical Example

Using Equation (B.26), we can now compute the most likely (single) equilibria for each of the versions of our games we have specified above (Γ^A through Γ^D) and for different initial belief values.

First, we must choose values for all the parameters in the game, consistent with the assumptions we made (i.e. $W_1 > W_4 > W_3, W_2 > W_4 > W_3$, and $W_1 > W_2$), so as to identify the numerical equilibria for each game when $\lambda = 1$. Furthermore, we center the game at zero by letting $g = b$ (see Section 3).

Since $W_4 > W_3$ and $W_4 \equiv g - b = 0$, $W_3 \equiv s - r < 0$. Let $s = 1$, $r = 2$ to fulfil this. Given those choices, $W_2 \equiv g - b + s = 1$, and $W_2 > W_4 > W_3$ holds. Further, let $t = 4$, such that $W_1 \equiv t - r = 2$, and $W_1 > W_4 > W_3$ as well as $W_1 > W_2$ is met. Finally, let us define the additional payoffs from monitoring (p, c) such that $W_3 < W_4$ is maintained. For ease of exposition, let $p = c = 0.25$ and to meet these conditions.

Given these parameter values, we can calculate the payoffs for each version of the game as follows:

- Γ^A : $W_1 = 2, W_2 = 1, W_3 = -1, W_4 = 0$.
- Γ^B : $W_1 = 2.25, W_2 = 1.25, W_3 = -0.75, W_4 = 0$.
- Γ^C : $W_1 = 2.25, W_2 = 1, W_3 = -0.75, W_4 = 0$.
- Γ^D : $W_1 = 2.5, W_2 = 1.25, W_3 = -0.5, W_4 = 0$.

Note that the choice of parameter values is, of course, somewhat arbitrary. As a result of our choice, as we will show in what follows, the mixed strategy Nash Equilibrium will

be moved closer towards the origin upon providing *private* benefits rather than *collective* ones. As we have shown in Appendix B.6, this does not have to be the case, but rather is an artifact of our choices for parameter values.

Equations (B.6), (B.12), (B.18), and (B.24) allow us to calculate the corresponding mixing probabilities pertaining to each of the MSNEs for the versions of our game given the numerical payoffs above.

- $\alpha_i^{*,A} = \frac{r-s}{t-2s} = \frac{2-1}{4-2(1)} = \frac{1}{2}$
- $\alpha_i^{*,B} = \frac{r-s-c}{t-2s-c} = \frac{2-1-0.25}{4-2(1)-0.25} = \frac{0.75}{1.75} = \frac{3}{7}$
- $\alpha_i^{*,C} = \frac{r-s-p}{t-2s} = \frac{2-1-0.25}{4-2(1)} = \frac{0.75}{2} = \frac{3}{8}$
- $\alpha_i^{*,D} = \frac{r-s-p-c}{t-2s-c} = \frac{2-1-0.25-0.25}{4-2(1)-0.25} = \frac{0.5}{1.75} = \frac{2}{7}$

Having identified these mixing probabilities, we must now choose an array of beliefs so as to see how optimal behavior varies when the assumption of $\lambda = 1$ is relaxed and beliefs about others' likelihood of cooperation begin to determine the outcome. To do this, we must choose values for these beliefs (β_j) so as to fall between the mixing probabilities identified above—in addition to values above (and below) the maximal (minimal) mixing probability, respectively—in order to demonstrate varying optimal behavior upon introducing monitoring parameters.

Recall Equation (B.27), and let $\beta_j = \frac{1}{4}$, such that $\beta_j < \alpha_i^{*,D} < \alpha_i^{*,C} < \alpha_i^{*,B} < \alpha_i^{*,A}$:

$$u_i^{\lambda=0}(\alpha_i, \beta_j = \frac{1}{4}) = \frac{\alpha_i W_1}{4} - \frac{\alpha_i W_3}{4} - \frac{\alpha_i W_2}{4} + \frac{\alpha_i W_4}{4} + \alpha_i W_3 - \alpha_i W_4 + \frac{W_2}{4} - \frac{W_4}{4} + W_4 \quad (\text{B.43})$$

Now apply payoffs from version Γ^A to determine the optimal strategy when $\lambda = 0$:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^A}(\alpha_i, \beta_j = \frac{1}{4}) &= \frac{2\alpha_i}{4} + \frac{\alpha_i}{4} - \frac{\alpha_i}{4} - \alpha_i + \frac{1}{4} \\ &= \frac{1 - 2\alpha_i}{4} \end{aligned} \quad (\text{B.44})$$

Which is maximized when $\alpha_i = 0$. Symmetrical initial beliefs of $\beta = \frac{1}{4}$ thus correspond to an optimal strategy of always defecting, (D, D) or W_4 , making it the only equilibrium.

Applying the payoffs from version Γ^B to Equation (B.43), we obtain:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^B}(\alpha_i, \beta_j = \frac{1}{4}) &= \frac{2.25\alpha_i}{4} + \frac{0.75\alpha_i}{4} - \frac{1.25\alpha_i}{4} - 0.75\alpha_i + \frac{1.25}{4} \\ &= \frac{1.25 - 1.25\alpha_i}{4} \end{aligned} \quad (\text{B.45})$$

Which is also maximized when $\alpha_i = 0$, making (D, D) or W_4 the only equilibrium.

Continuing with version Γ^C , we obtain:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^C}(\alpha_i, \beta_j = \frac{1}{4}) &= \frac{2.25\alpha_i}{4} + \frac{0.75\alpha_i}{4} - \frac{\alpha_i}{4} - 0.75\alpha_i + \frac{1}{4} \\ &= \frac{1 - \alpha_i}{4} \end{aligned} \quad (\text{B.46})$$

Which again is maximized when $\alpha_i = 0$.

Turning to the fully augmented version of our game, Γ^D , we obtain:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^D}(\alpha_i, \beta_j = \frac{1}{4}) &= \frac{2.5\alpha_i}{4} + \frac{0.5\alpha_i}{4} - \frac{1.25\alpha_i}{4} - 0.5\alpha_i + \frac{1.25}{4} \\ &= \frac{1.25 - 0.25\alpha_i}{4} \end{aligned} \quad (\text{B.47})$$

As expected, since beliefs (β_j) are still smaller than the version's optimal mixing probabilities under the game's MSNE, this equation, too, is maximized when $\alpha_i = 0$. When the beliefs in others' cooperation are low (e.g., $\beta_j = \frac{1}{4}$), all versions point towards uniform defection by the actors when beliefs (and not rationality) inform the actors' choices.

Now, return to Equation (B.27), and let $\beta_j = \frac{1}{3}$, such that $\alpha_i^{*,D} < \beta_j < \alpha_i^{*,C} < \alpha_i^{*,B} < \alpha_i^{*,A}$:

$$u_i^{\lambda=0}(\alpha_i, \beta_j = \frac{1}{3}) = \frac{\alpha_i W_1}{3} - \frac{\alpha_i W_3}{3} - \frac{\alpha_i W_2}{3} + \frac{\alpha_i W_4}{3} + \alpha_i W_3 - \alpha_i W_4 + \frac{W_2}{3} - \frac{W_4}{3} + W_4 \quad (\text{B.48})$$

Again, we begin by plugging in the payoffs of version Γ^A to determine the optimal strategy when $\lambda = 0$:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^A}(\alpha_i, \beta_j = \frac{1}{3}) &= \frac{2\alpha_i}{3} + \frac{\alpha_i}{3} - \frac{\alpha_i}{3} - \alpha_i + \frac{1}{3} \\ &= \frac{1 - \alpha_i}{3} \end{aligned} \quad (\text{B.49})$$

Which is maximized when $\alpha_i = 0$, thus yielding uniform defection as the optimal strategy for player i , and also for all players j due to payoff symmetry.

Applying the payoffs from version Γ^B to Equation (B.48), we obtain:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^B}(\alpha_i, \beta_j = \frac{1}{3}) &= \frac{2.25\alpha_i}{3} + \frac{0.75\alpha_i}{3} - \frac{1.25\alpha_i}{3} - 0.75\alpha_i + \frac{1.25}{3} \\ &= \frac{5 - 2\alpha_i}{12} \end{aligned} \quad (\text{B.50})$$

Which is also maximized when $\alpha_i = 0$, and the logic above with respect to optimal strategy continues to apply.

Turning to version Γ^C , we obtain:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^C}(\alpha_i, \beta_j = \frac{1}{3}) &= \frac{2.25\alpha_i}{3} + \frac{0.75\alpha_i}{3} - \frac{\alpha_i}{3} - 0.75\alpha_i + \frac{1}{3} \\ &= \frac{5 - \alpha_i}{12} \end{aligned} \quad (\text{B.51})$$

Which again is maximized when $\alpha_i = 0$, yielding (D, D) or W_4 as the only equilibrium in

the stability set when $\beta_j = \frac{1}{3}$.

Turning to the fully augmented version of our game, Γ^D , we obtain:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^D}(\alpha_i, \beta_j = \frac{1}{3}) &= \frac{2.5\alpha_i}{3} + \frac{0.5\alpha_i}{3} - \frac{1.25\alpha_i}{3} - 0.5\alpha_i + \frac{1.25}{3} \\ &= \frac{5 + \alpha_i}{12} \end{aligned} \quad (\text{B.52})$$

Which conversely is maximized when $\alpha_i = 1$. Consequently, an equilibrium of (C, C) or W_1 is in the stability set: when beliefs regarding the likelihood of cooperation of other actors are slightly larger (i.e. $\beta_j = \frac{1}{3}$ rather than $\beta_j = \frac{1}{4}$), we see varying equilibria in the stability sets for version Γ^D , holding all other factors constant.

Again returning to Equation (B.27), now let $\beta_j = \frac{2}{5}$, i.e., $\alpha_i^{*,D} < \alpha_i^{*,C} < \beta_j < \alpha_i^{*,B} < \alpha_i^{*,A}$:

$$u_i^{\lambda=0}(\alpha_i, \beta_j = \frac{2}{5}) = \frac{2\alpha_i W_1}{5} - \frac{2\alpha_i W_3}{5} - \frac{2\alpha_i W_2}{5} + \frac{2\alpha_i W_4}{5} + \alpha_i W_3 - \alpha_i W_4 + \frac{2W_2}{5} - \frac{2W_4}{5} + W_4 \quad (\text{B.53})$$

Now, apply payoffs from version Γ^A to determine the optimal strategy when $\lambda = 0$:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^A}(\alpha_i, \beta_j = \frac{2}{5}) &= \frac{4\alpha_i}{5} + \frac{2\alpha_i}{5} - \frac{2\alpha_i}{5} - \alpha_i + \frac{2}{5} \\ &= \frac{2 - \alpha_i}{5} \end{aligned} \quad (\text{B.54})$$

Which is maximized when $\alpha_i = 0$, making (D, D) or W_4 in the stability set.

Applying the payoffs from version Γ^B to Equation (B.53), we obtain:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^B}(\alpha_i, \beta_j = \frac{2}{5}) &= \frac{4.5\alpha_i}{5} + \frac{1.5\alpha_i}{5} - \frac{2.5\alpha_i}{5} - 0.75\alpha_i + \frac{2.5}{5} \\ &= \frac{5 - 0.5\alpha_i}{10} \end{aligned} \quad (\text{B.55})$$

Which is also still maximized when $\alpha_i = 0$.

As for version Γ^C , we obtain:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^C}(\alpha_i, \beta_j = \frac{2}{5}) &= \frac{4.5\alpha_i}{5} + \frac{1.5\alpha_i}{5} - \frac{2\alpha_i}{5} - 0.75\alpha_i + \frac{2}{5} \\ &= \frac{4 + 0.5\alpha_i}{10} \end{aligned} \quad (\text{B.56})$$

Which is now maximized when $\alpha_i = 1$, thus no longer yielding the uncooperative equilibrium in its stability set.

Similarly, for the fully augmented version of our game, Γ^D , we obtain:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^D}(\alpha_i, \beta_j = \frac{2}{5}) &= \frac{5\alpha_i}{5} + \frac{\alpha_i}{5} - \frac{2.5\alpha_i}{5} - 0.5\alpha_i + \frac{2.5}{5} \\ &= \frac{2.5 + \alpha_i}{5} \end{aligned} \quad (\text{B.57})$$

Again considering Equation (B.27), now let $\beta_j = \frac{4}{9}$, i.e., $\alpha_i^{*,D} < \alpha_i^{*,C} < \alpha_i^{*,B} < \beta_j < \alpha_i^{*,A}$:

$$u_i^{\lambda=0}(\alpha_i, \beta_j = \frac{4}{9}) = \frac{4\alpha_i W_1}{9} - \frac{4\alpha_i W_3}{9} - \frac{4\alpha_i W_2}{9} + \frac{4\alpha_i W_4}{9} + \alpha_i W_3 - \alpha_i W_4 + \frac{4W_2}{9} - \frac{4W_4}{9} + W_4 \quad (\text{B.58})$$

Now apply payoffs from version Γ^A to determine the optimal strategy when $\lambda = 0$:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^A}(\alpha_i, \beta_j = \frac{4}{9}) &= \frac{8\alpha_i}{9} + \frac{4\alpha_i}{9} - \frac{4\alpha_i}{9} - \alpha_i + \frac{4}{9} \\ &= \frac{4 - \alpha_i}{9} \end{aligned} \quad (\text{B.59})$$

Which remains optimized at $\alpha_i = 0$, and thus only holds (D, D) or W_1 in the stability set.

Applying the payoffs from version Γ^B to Equation (B.58), we obtain:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^B}(\alpha_i, \beta_j = \frac{4}{9}) &= \frac{9\alpha_i}{9} + \frac{3\alpha_i}{9} - \frac{5\alpha_i}{9} - 0.75\alpha_i + \frac{5}{9} \\ &= \frac{20 + \alpha_i}{36} \end{aligned} \quad (\text{B.60})$$

Which is now optimized at $\alpha_i = 1$, thus yielding the cooperative equilibrium in the stability set.

Continuing with version Γ^C , we obtain:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^C}(\alpha_i, \beta_j = \frac{4}{9}) &= \frac{9\alpha_i}{9} + \frac{3\alpha_i}{9} - \frac{4\alpha_i}{9} - 0.75\alpha_i + \frac{4}{9} \\ &= \frac{16 + 5\alpha_i}{36} \end{aligned} \quad (\text{B.61})$$

Which can be optimized when $\alpha_i = 1$.

Turning to the fully augmented version of our game, Γ^D , we obtain:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^D}(\alpha_i, \beta_j = \frac{4}{9}) &= \frac{10\alpha_i}{9} + \frac{2\alpha_i}{9} - \frac{5\alpha_i}{9} - 0.5\alpha_i + \frac{5}{9} \\ &= \frac{10 + 5\alpha_i}{18} \end{aligned} \quad (\text{B.62})$$

Which again yields the cooperative equilibrium (C, C) or W_1 as the sole equilibrium in the stability set.

Lastly, consider Equation (B.27), and let $\beta_j = \frac{2}{3}$, i.e., $\alpha_i^{*,D} < \alpha_i^{*,C} < \alpha_i^{*,B} < \alpha_i^{*,A} < \beta_j$:

$$u_i^{\lambda=0}(\alpha_i, \beta_j = \frac{2}{3}) = \frac{2\alpha_i W_1}{3} - \frac{2\alpha_i W_3}{3} - \frac{2\alpha_i W_2}{3} + \frac{2\alpha_i W_4}{3} + \alpha_i W_3 - \alpha_i W_4 + \frac{2W_2}{3} - \frac{2W_4}{3} + W_4 \quad (\text{B.63})$$

Now apply payoffs from version Γ^A to determine the optimal strategy when $\lambda = 0$:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^A}(\alpha_i, \beta_j = \frac{2}{3}) &= \frac{4\alpha_i}{3} + \frac{2\alpha_i}{3} - \frac{2\alpha_i}{3} - \alpha_i + \frac{2}{3} \\ &= \frac{2 + \alpha_i}{3} \end{aligned} \quad (\text{B.64})$$

Which now holds the cooperative equilibrium in the stability set, as it is maximized when $\alpha_i = 1$.

Applying the payoffs from version Γ^B to Equation (B.63), we obtain:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^B}(\alpha_i, \beta_j = \frac{2}{3}) &= \frac{4.5\alpha_i}{3} + \frac{1.5\alpha_i}{3} - \frac{2.5\alpha_i}{3} - 0.75\alpha_i + \frac{2.25}{3} \\ &= \frac{9 + 5\alpha_i}{12} \end{aligned} \quad (\text{B.65})$$

Which too is maximized when $\alpha_i = 1$.

Continuing with version Γ^C , we obtain:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^C}(\alpha_i, \beta_j = \frac{2}{3}) &= \frac{4.5\alpha_i}{3} + \frac{1.5\alpha_i}{3} - \frac{2\alpha_i}{3} - 0.75\alpha_i + \frac{2}{3} \\ &= \frac{8 + 7\alpha_i}{12} \end{aligned} \quad (\text{B.66})$$

Finally, turning to the fully augmented version of our game, Γ^D , we obtain:

$$\begin{aligned} u_i^{\lambda=0, \Gamma^D}(\alpha_i, \beta_j = \frac{2}{3}) &= \frac{5\alpha_i}{3} + \frac{1\alpha_i}{3} - \frac{2.5\alpha_i}{3} - 0.5\alpha_i + \frac{2.5}{3} \\ &= \frac{2.5 + 2\alpha_i}{3} \end{aligned} \quad (\text{B.67})$$

As we can see, when beliefs are sufficiently large, all versions of our games discussed here (i.e., Γ^A through Γ^D) hold the cooperative equilibrium (C, C) or W_1 in the stability set. If beliefs are a random variable, it becomes evident that versions that include monitoring or protection (i.e., Γ^B through Γ^D) have cooperative equilibria in the stability set for a greater

portion of the probability density function (pdf) of the beliefs. Thus, regardless of the shape of the pdf, cooperative outcomes become more likely upon introducing monitoring (c) and/or protection (p) parameters.

Appendix C Comparative Statics

Recall from Appendix (B.4.2) that the augmented version of our game's (Γ^D) mixed strategy Nash Equilibrium is given by:

$$\alpha_i^* = \frac{g - b + r - s - c - p}{t - 2s - c}$$

Further recall that by virtue of the consistently applied assumption of the assurance game (or Stag Hunt) for versions Γ^A through Γ^D are:

$$t - r > g - b + s > g - b > s - r$$

and

$$t - r + c + p > g - b + s + c > g - b > s - r + p + c$$

and all of the above parameters are assumed non-negative. Against this backdrop, we can compute comparative statics.

C.1 Collective Benefits from Monitoring, c

$$\begin{aligned} \frac{\partial \alpha_i^*}{\partial c} &= \frac{r - s - c - p}{(t - 2s - c)^2} - \frac{1}{t - 2s - c} \\ &= \frac{r - s - c - p}{(t - 2s - c)^2} - \frac{t - 2s - c}{(t - 2s - c)^2} \\ &= \frac{r - s - c - p - t + 2s + c}{(t - 2s - c)^2} \\ &= \frac{r - t + s - p}{(t - 2s - c)^2} \end{aligned}$$

Which is negative, since $t - r + p > s$, as shown in Appendix B.4.1. Hence, an increase in collective benefits emanating from monitoring coincides with a shrinking MSNE, and thus

greater likelihood of cooperation for any given probability distribution of prior beliefs.

C.2 Private Benefits from Monitoring, p

$$\frac{\partial \alpha_i^*}{\partial p} = -\frac{1}{t - 2s - c}$$

The partial derivative of α_i^* is negative, as $t > 2s + c$, which is necessary for the existence of the MSNE (see Appendix B.4.2). Hence, an increase in the private benefits emanating from monitoring coincides with a shrinking MSNE, and thus greater likelihood of cooperation for any given probability distribution of prior beliefs.

C.3 Bureaucrat Retaliation, r

$$\frac{\partial \alpha_i^*}{\partial r} = \frac{1}{t - 2s - c}$$

Again, the partial derivative of α_i^* is positive, as $t > 2s + c$, which is necessary for the existence of the MSNE (see Appendix B.4.2). As the MSNE shifts towards one, and given a probability distribution of prior beliefs, taking costly action against corruption becomes less likely.

As above, we assume that $t - r > g - b + s$, such that the cost from potential retaliation, r , cannot exceed the difference between joint cooperation and singular cooperation, $t - s$. If r did increase beyond that threshold, the game ceases to be an assurance game, and takes on a prisoner's dilemma-type of structure (see Table D.1 in Appendix D). As described above, such a move is theoretically and substantively illogical.

C.4 Benefits from Joint Cooperation, t

$$\frac{\partial \alpha_i^*}{\partial t} = -\frac{r - s - c - p}{(t - 2s - c)^2}$$

Which is negative, because $r > s + p + c$ by assumption, as shown in Appendix B.4.2. Consequently, increases in these benefits shift the MSNE towards zero, and under any given probability distribution of initial beliefs, coincide with taking more costly action against corruption.

C.5 Benefits from Individual Cooperation, s

$$\begin{aligned} \frac{\partial \alpha_i^*}{\partial s} &= \frac{2(r - s - c - p)}{(t - 2s - c)^2} - \frac{1}{t - 2s - c} \\ &= \frac{2(r - s - c - p)}{(t - 2s - c)^2} - \frac{t - 2s - c}{(t - 2s - c)^2} \\ &= \frac{2r - 2s - 2c - 2p - t + 2s + c}{(t - 2s - c)^2} \\ &= \frac{2r - c - 2p - t}{(t - 2s - c)^2} \end{aligned}$$

Which can be either positive if $(t > 2r - c - 2p)$ or negative otherwise. In other words, if benefits from joint cooperation (t) are sufficiently large, increases to s shift the MSNE towards one, thus rendering cooperation less likely—i.e., under a given probability distribution of initial beliefs. This might seem counterintuitive at first, but recall that these benefits accrue to free-riding actors as much as they do to individual cooperators.

Since we assume $r > s + p + c$ and $t > s + r + c$ to maintain the assurance game, as shown in Appendix B.4.2, s can only increase to a limited degree, *ceteris paribus*, before the game no longer resembles an assurance game.

Appendix D Basic Collective Action Games

Table D.1: Types of Collective Action Games

		<i>Citizen 2</i>		<i>Citizen 2</i>	
		<i>Cooperate/ Take Action</i>	<i>Defect/ Do Nothing</i>	<i>Cooperate/ Take Action</i>	<i>Defect/ Do Nothing</i>
		Prisoner's Dilemma		Assurance Game/Stag Hunt	
<i>Citizen 1</i>	<i>Cooperate/ Take Action</i>	2, 2	0, 3	3, 3	0, 2
	<i>Defect/ Do Nothing</i>	3, 0	1, 1	2, 0	1, 1
		Deadlock		Harmony	
<i>Citizen 1</i>	<i>Cooperate/ Take Action</i>	0, 0	1, 2	3, 3	2, 1
	<i>Defect/ Do Nothing</i>	2, 1	3, 3	1, 2	0, 0
		Chicken/Hawk-Dove			
<i>Citizen 1</i>	<i>Cooperate/ Take Action</i>	2, 2	1, 3		
	<i>Defect/ Do Nothing</i>	3, 1	0, 0		

Note: The numeric payoffs denote the preference orderings in each game, which are not perfectly comparable across games. For more, see [Dixit, Skeath and Reiley \(2014\)](#) and [Humphreys \(2017\)](#).